

Space Invaders

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Vector2D.java	106

Chapter 2

Namespace Index

2.1 Namespace List

Here is a list of all namespaces with brief descriptions:

en.tresz.spaceinvaders	19
en.tresz.spaceinvaders.game	19
en.tresz.spaceinvaders.game.objects	20
en.tresz.spaceinvaders.game.objects.aliens	20
en.tresz.spaceinvaders.game.objects.projectiles	20
en.tresz.spaceinvaders.menu	20
en.tresz.spaceinvaders.util	21

Chapter 3

Hierarchical Index

3.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

ActionListener	
en.tresz.spaceinvaders.GameOverWindow	46
en.tresz.spaceinvaders.HardGameOverWindow	55
en.tresz.spaceinvaders.game.AlienController	28
en.tresz.spaceinvaders.util.ButtonMaker	33
en.tresz.spaceinvaders.game.objects.GameObject	42
en.tresz.spaceinvaders.game.objects.Player	69
en.tresz.spaceinvaders.game.objects.aliens.Alien	23
en.tresz.spaceinvaders.game.objects.aliens.FastAlien	36
en.tresz.spaceinvaders.game.objects.aliens.FastRapidfireAlien	39
en.tresz.spaceinvaders.game.objects.aliens.NormalAlien	66
en.tresz.spaceinvaders.game.objects.aliens.RapidfireAlien	78
en.tresz.spaceinvaders.game.objects.aliens.WideAlien	91
en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile	31
en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile	75
en.tresz.spaceinvaders.util.ImageLoader	60
JComponent	
en.tresz.spaceinvaders.util.HealthBar	58
JFrame	
en.tresz.spaceinvaders.GameOverWindow	46
en.tresz.spaceinvaders.HardGameOverWindow	55
en.tresz.spaceinvaders.MainWindow	63
JLabel	
en.tresz.spaceinvaders.util.GameTimer	53
JPanel	
en.tresz.spaceinvaders.game.GamePanel	48
en.tresz.spaceinvaders.menu.DifficultyMenuPanel	34
en.tresz.spaceinvaders.menu.MainMenuPanel	62
en.tresz.spaceinvaders.menu.ScoreboardPanel	82
en.tresz.spaceinvaders.Main	61
en.tresz.spaceinvaders.MainWindow.PanelType	69
en.tresz.spaceinvaders.game.PlayerController	74
en.tresz.spaceinvaders.util.Score	81
en.tresz.spaceinvaders.util.ScoreManager	83
en.tresz.spaceinvaders.util.Time	85

en.tresz.spaceinvaders.util.Vector2D	87
WindowListener	
en.tresz.spaceinvaders.GameOverWindow	46
en.tresz.spaceinvaders.HardGameOverWindow	55

Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

en.tresz.spaceinvaders.game.objects.aliens.Alien	23
The Alien class represents a generic alien	
en.tresz.spaceinvaders.game.AlienController	28
Manages alien collective behavior including movement, collisions, and destruction	
en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile	31
A projectile fired by aliens moving downwards	
en.tresz.spaceinvaders.util.ButtonMaker	33
A utility class to create customized buttons	
en.tresz.spaceinvaders.menu.DifficultyMenuPanel	34
The difficulty selection screen	
en.tresz.spaceinvaders.game.objects.aliens.FastAlien	36
A fast-moving alien that moves three times faster than normal aliens	
en.tresz.spaceinvaders.game.objects.aliens.FastRapidfireAlien	39
A fast-moving alien that also shoots three times faster than normal	
en.tresz.spaceinvaders.game.objects.GameObject	42
The base class for all game objects	
en.tresz.spaceinvaders.GameOverWindow	46
Game over window for easy/medium difficulty with play again option	
en.tresz.spaceinvaders.game.GamePanel	48
The panel where the game runs	
en.tresz.spaceinvaders.util.GameTimer	53
A label that displays and tracks game time in MM:SS format	
en.tresz.spaceinvaders.HardGameOverWindow	55
Game over window for hard difficulty that allows entering player name for scoreboard	
en.tresz.spaceinvaders.util.HealthBar	58
A visual health bar component that displays health as heart images	
en.tresz.spaceinvaders.util.ImageLoader	60
A utility class to safely load images from the resources folder	
en.tresz.spaceinvaders.Main	61
The entry point of the Space Invaders application	
en.tresz.spaceinvaders.menu.MainMenuPanel	62
The main menu panel with options to play, view scoreboard, or exit	
en.tresz.spaceinvaders.MainWindow	63
The main frame	
en.tresz.spaceinvaders.game.objects.aliens.NormalAlien	66
A standard alien with normal movement and shooting patterns	

en.tresz.spaceinvaders.MainWindow.PanelType	69
en.tresz.spaceinvaders.game.objects.Player	
Represents the player's spaceship controlled by keyboard input	69
en.tresz.spaceinvaders.game.PlayerController	
Handles player-related game logic including collision detection with alien projectiles	74
en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile	
A projectile fired by the player moving upwards	75
en.tresz.spaceinvaders.game.objects.aliens.RapidfireAlien	
An alien that shoots three times faster than normal aliens	78
en.tresz.spaceinvaders.util.Score	
Represents a player's score with their name and completion time	81
en.tresz.spaceinvaders.menu.ScoreboardPanel	
The scoreboard panel displays the top 5 high scores	82
en.tresz.spaceinvaders.util.ScoreManager	
Manages saving and loading of high scores using JSON serialization	83
en.tresz.spaceinvaders.util.Time	
Represents a time duration in minutes and seconds	85
en.tresz.spaceinvaders.util.Vector2D	
2D vector class for positioning	87
en.tresz.spaceinvaders.game.objects.aliens.WideAlien	
A wide alien that shoots three projectiles simultaneously	91

Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

src/main/java/en/tresz/spaceinvaders/GameOverWindow.java	100
src/main/java/en/tresz/spaceinvaders/HardGameOverWindow.java	101
src/main/java/en/tresz/spaceinvaders/Main.java	101
src/main/java/en/tresz/spaceinvaders/MainWindow.java	101
src/main/java/en/tresz/spaceinvaders/game/AlienController.java	95
src/main/java/en/tresz/spaceinvaders/game/GamePanel.java	95
src/main/java/en/tresz/spaceinvaders/game/PlayerController.java	100
src/main/java/en/tresz/spaceinvaders/game/objects/GameObject.java	98
src/main/java/en/tresz/spaceinvaders/game/objects/Player.java	99
src/main/java/en/tresz/spaceinvaders/game/objects/aliens/Alien.java	96
src/main/java/en/tresz/spaceinvaders/game/objects/aliens/FastAlien.java	96
src/main/java/en/tresz/spaceinvaders/game/objects/aliens/FastRapidfireAlien.java	97
src/main/java/en/tresz/spaceinvaders/game/objects/aliens/NormalAlien.java	97
src/main/java/en/tresz/spaceinvaders/game/objects/aliens/RapidfireAlien.java	97
src/main/java/en/tresz/spaceinvaders/game/objects/aliens/WideAlien.java	98
src/main/java/en/tresz/spaceinvaders/game/objects/projectiles/AlienProjectile.java	99
src/main/java/en/tresz/spaceinvaders/game/objects/projectiles/PlayerProjectile.java	99
src/main/java/en/tresz/spaceinvaders/menu/DifficultyMenuPanel.java	102
src/main/java/en/tresz/spaceinvaders/menu/MainMenuPanel.java	102
src/main/java/en/tresz/spaceinvaders/menu/ScoreboardPanel.java	103
src/main/java/en/tresz/spaceinvaders/util/ButtonMaker.java	104
src/main/java/en/tresz/spaceinvaders/util/GameTimer.java	104
src/main/java/en/tresz/spaceinvaders/util/HealthBar.java	104
src/main/java/en/tresz/spaceinvaders/util/ImageLoader.java	105
src/main/java/en/tresz/spaceinvaders/util/Score.java	105
src/main/java/en/tresz/spaceinvaders/util/ScoreManager.java	105
src/main/java/en/tresz/spaceinvaders/util/Time.java	106
src/main/java/en/tresz/spaceinvaders/util/Vector2D.java	106

Chapter 6

Directory Documentation

6.1 `src/main/java/en/tresz/spaceinvaders/game/objects/aliens` Directory Reference

Files

- file [Alien.java](#)
- file [FastAlien.java](#)
- file [FastRapidfireAlien.java](#)
- file [NormalAlien.java](#)
- file [RapidfireAlien.java](#)
- file [WideAlien.java](#)

6.2 `src/main/java/en` Directory Reference

Directories

- directory [tresz](#)

6.3 `src/main/java/en/tresz/spaceinvaders/game` Directory Reference

Directories

- directory [objects](#)

Files

- file [AlienController.java](#)
- file [GamePanel.java](#)
- file [PlayerController.java](#)

6.4 src/main/java Directory Reference

Directories

- directory [en](#)

6.5 src/main Directory Reference

Directories

- directory [java](#)

6.6 src/main/java/en/tresz/spaceinvaders/menu Directory Reference

Files

- file [DifficultyMenuPanel.java](#)
- file [MainMenuPanel.java](#)
- file [ScoreboardPanel.java](#)

6.7 src/main/java/en/tresz/spaceinvaders/game/objects Directory Reference

Directories

- directory [aliens](#)
- directory [projectiles](#)

Files

- file [GameObject.java](#)
- file [Player.java](#)

6.8 src/main/java/en/tresz/spaceinvaders/game/objects/projectiles Directory Reference

Files

- file [AlienProjectile.java](#)
- file [PlayerProjectile.java](#)

6.9 src/main/java/en/tresz/spaceinvaders Directory Reference

Directories

- directory [game](#)
- directory [menu](#)
- directory [util](#)

Files

- file [GameOverWindow.java](#)
- file [HardGameOverWindow.java](#)
- file [Main.java](#)
- file [MainWindow.java](#)

6.10 src Directory Reference

Directories

- directory [main](#)

6.11 src/main/java/en/tresz Directory Reference

Directories

- directory [spaceinvaders](#)

6.12 src/main/java/en/tresz/spaceinvaders/util Directory Reference

Files

- file [ButtonMaker.java](#)
- file [GameTimer.java](#)
- file [HealthBar.java](#)
- file [ImageLoader.java](#)
- file [Score.java](#)
- file [ScoreManager.java](#)
- file [Time.java](#)
- file [Vector2D.java](#)

Chapter 7

Namespace Documentation

7.1 Package en.tresz.spaceinvaders

Packages

- package [game](#)
- package [menu](#)
- package [util](#)

Classes

- class [GameOverWindow](#)
Game over window for easy/medium difficulty with play again option.
- class [HardGameOverWindow](#)
Game over window for hard difficulty that allows entering player name for scoreboard.
- class [Main](#)
The entry point of the Space Invaders application.
- class [MainWindow](#)
The main frame.

7.2 Package en.tresz.spaceinvaders.game

Packages

- package [objects](#)

Classes

- class [AlienController](#)
Manages alien collective behavior including movement, collisions, and destruction.
- class [GamePanel](#)
The panel where the game runs.
- class [PlayerController](#)
Handles player-related game logic including collision detection with alien projectiles.

7.3 Package en.tresz.spaceinvaders.game.objects

Packages

- package [aliens](#)
- package [projectiles](#)

Classes

- class [GameObject](#)
The base class for all game objects.
- class [Player](#)
Represents the player's spaceship controlled by keyboard input.

7.4 Package en.tresz.spaceinvaders.game.objects.aliens

Classes

- class [Alien](#)
The [Alien](#) class represents a generic alien.
- class [FastAlien](#)
A fast-moving alien that moves three times faster than normal aliens.
- class [FastRapidfireAlien](#)
A fast-moving alien that also shoots three times faster than normal.
- class [NormalAlien](#)
A standard alien with normal movement and shooting patterns.
- class [RapidfireAlien](#)
An alien that shoots three times faster than normal aliens.
- class [WideAlien](#)
A wide alien that shoots three projectiles simultaneously.

7.5 Package en.tresz.spaceinvaders.game.objects.projectiles

Classes

- class [AlienProjectile](#)
A projectile fired by aliens moving downwards.
- class [PlayerProjectile](#)
A projectile fired by the player moving upwards.

7.6 Package en.tresz.spaceinvaders.menu

Classes

- class [DifficultyMenuPanel](#)
The difficulty selection screen.
- class [MainMenuPanel](#)
The main menu panel with options to play, view scoreboard, or exit.
- class [ScoreboardPanel](#)
The scoreboard panel displays the top 5 high scores.

7.7 Package en.tresz.spaceinvaders.util

Classes

- class [ButtonMaker](#)
A utility class to create customized buttons.
- class [GameTimer](#)
A label that displays and tracks game time in MM:SS format.
- class [HealthBar](#)
A visual health bar component that displays health as heart images.
- class [ImageLoader](#)
A utility class to safely load images from the resources folder.
- class [Score](#)
Represents a player's score with their name and completion time.
- class [ScoreManager](#)
Manages saving and loading of high scores using JSON serialization.
- class [Time](#)
Represents a time duration in minutes and seconds.
- class [Vector2D](#)
2D vector class for positioning.

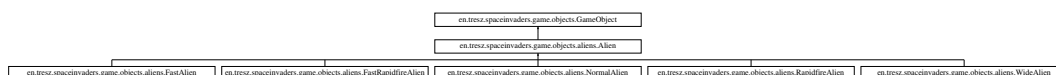
Chapter 8

Class Documentation

8.1 en.tresz.spaceinvaders.game.objects.aliens.Alien Class Reference

The [Alien](#) class represents a generic alien.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.aliens.Alien:



Public Member Functions

- void [update](#) ([GamePanel](#) gamePanel)
Updates the alien's position and handles wall collisions.
- void [shoot](#) ([GamePanel](#) gamePanel)
Attempts to shoot a projectile towards the player.
- void [draw](#) ([Graphics](#) g)
Draws the alien on the screen using its image.
- boolean [reachedBottom](#) ([Player](#) player)
Checks if the alien has reached the player's vertical position.
- boolean [isRequestGlobalMoveDown](#) ()
Checks if the alien requests all aliens to move down, resetting the flag after checking.
- int [getClosingInTreshold](#) ()
Gets the collision threshold before the alien requests a global move down.
- void [setCollisionCount](#) (int count)
Sets the collision count for this alien.
- int [getHitSpeed](#) ()
Gets the hit speed multiplier for this alien.
- boolean [chance](#) (double percentage)
Determines if a random event should occur based on a given percentage.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Static Public Attributes

- static final int [ALIEN_HEIGHT](#) = 80

Protected Member Functions

- [Alien](#) ([Alien](#) alien)
Copy constructor for creating a new [Alien](#) from an existing one.
- [Alien](#) ([Vector2D](#) p, [Vector2D](#) v, int width, int hitSpeed)
Constructs a new [Alien](#) with the specified parameters.

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes

- [BufferedImage](#) [alienImage](#)
- int [collisionCount](#) = 0
- int [closingInTreshold](#) = 6
- int [hitSpeed](#)
- boolean [requestGlobalMoveDown](#) = false
- int [shootingInterval](#) = 500

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [Vector2D](#) *position*
- [Vector2D](#) *velocity*
- int *width*
- int *height*

8.1.1 Detailed Description

The [Alien](#) class represents a generic alien.

8.1.2 Constructor & Destructor Documentation

8.1.2.1 Alien() [1/2]

```
en.tresz.spaceinvaders.game.objects.aliens.Alien.Alien (
    Alien alien) [inline], [protected]
```

Copy constructor for creating a new [Alien](#) from an existing one.

Parameters

<i>alien</i>	the alien to copy from
--------------	------------------------

8.1.2.2 Alien() [2/2]

```
en.tresz.spaceinvaders.game.objects.aliens.Alien.Alien (
    Vector2D p,
    Vector2D v,
    int width,
    int hitSpeed) [inline], [protected]
```

Constructs a new [Alien](#) with the specified parameters.

Parameters

<i>p</i>	the initial position of the alien
<i>v</i>	the initial velocity of the alien
<i>width</i>	the width of the alien
<i>hitSpeed</i>	the speed multiplier for movement and shooting

8.1.3 Member Function Documentation

8.1.3.1 chance()

```
boolean en.tresz.spaceinvaders.game.objects.aliens.Alien.chance (
    double percentage) [inline]
```

Determines if a random event should occur based on a given percentage.

Parameters

<i>percentage</i>	the probability of the event occurring (0-100)
-------------------	--

Returns

true if the random event occurs, false otherwise

8.1.3.2 draw()

```
void en.tresz.spaceinvaders.game.objects.aliens.Alien.draw (
    Graphics g) [inline]
```

Draws the alien on the screen using its image.

Parameters

<i>g</i>	the graphics context to draw on
----------	---------------------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

8.1.3.3 getClosingInTreshold()

```
int en.tresz.spaceinvaders.game.objects.aliens.Alien.getClosingInTreshold () [inline]
```

Gets the collision threshold before the alien requests a global move down.

Returns

the closing in threshold value

8.1.3.4 getHitSpeed()

```
int en.tresz.spaceinvaders.game.objects.aliens.Alien.getHitSpeed () [inline]
```

Gets the hit speed multiplier for this alien.

Returns

the hit speed value

8.1.3.5 isRequestGlobalMoveDown()

```
boolean en.tresz.spaceinvaders.game.objects.aliens.Alien.isRequestGlobalMoveDown () [inline]
```

Checks if the alien requests all aliens to move down, resetting the flag after checking.

Returns

true if the alien requests all aliens to move down, false otherwise

8.1.3.6 reachedBottom()

```
boolean en.tresz.spaceinvaders.game.objects.aliens.Alien.reachedBottom (
    Player player) [inline]
```

Checks if the alien has reached the player's vertical position.

Parameters

<i>player</i>	the player to check against
---------------	-----------------------------

Returns

true if the alien has reached the player's level, false otherwise

8.1.3.7 setCollisionCount()

```
void en.tresz.spaceinvaders.game.objects.aliens.Alien.setCollisionCount (  
    int count) [inline]
```

Sets the collision count for this alien.

Parameters

<i>count</i>	the new collision count value
--------------	-------------------------------

8.1.3.8 shoot()

```
void en.tresz.spaceinvaders.game.objects.aliens.Alien.shoot (  
    GamePanel gamePanel) [inline]
```

Attempts to shoot a projectile towards the player.

The shooting interval is controlled by hitSpeed and resets after each shot.

Parameters

<i>gamePanel</i>	the game panel to add the projectile to
------------------	---

Reimplemented in [en.tresz.spaceinvaders.game.objects.aliens.WideAlien](#).

8.1.3.9 update()

```
void en.tresz.spaceinvaders.game.objects.aliens.Alien.update (  
    GamePanel gamePanel) [inline]
```

Updates the alien's position and handles wall collisions.

Increments collision count when hitting walls and requests global move down when threshold is reached.

Parameters

<i>gamePanel</i>	the game panel for boundary checking
------------------	--------------------------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

8.1.4 Member Data Documentation

8.1.4.1 ALIEN_HEIGHT

```
final int en.tresz.spaceinvaders.game.objects.aliens.Alien.ALIEN_HEIGHT = 80 [static]
```

8.1.4.2 alienImage

```
BufferedImage en.tresz.spaceinvaders.game.objects.aliens.Alien.alienImage [protected]
```

8.1.4.3 closingInTreshold

```
int en.tresz.spaceinvaders.game.objects.aliens.Alien.closingInTreshold = 6 [protected]
```

8.1.4.4 collisionCount

```
int en.tresz.spaceinvaders.game.objects.aliens.Alien.collisionCount = 0 [protected]
```

8.1.4.5 hitSpeed

```
int en.tresz.spaceinvaders.game.objects.aliens.Alien.hitSpeed [protected]
```

8.1.4.6 requestGlobalMoveDown

```
boolean en.tresz.spaceinvaders.game.objects.aliens.Alien.requestGlobalMoveDown = false [protected]
```

8.1.4.7 shootingInterval

```
int en.tresz.spaceinvaders.game.objects.aliens.Alien.shootingInterval = 500 [protected]
```

The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/objects/aliens/Alien.java](#)

8.2 en.tresz.spaceinvaders.game.AlienController Class Reference

Manages alien collective behavior including movement, collisions, and destruction.

Public Member Functions

- void [moveDownAllAliens](#) ([GamePanel](#) gamePanel)
Moves all aliens down by one row height.
- void [handleAllAlienCollisions](#) ([GamePanel](#) gamePanel)
Handles collisions between all aliens, reversing velocities and separating overlapping aliens.
- void [addAllAliens](#) (List< [GameObject](#) > [objects](#), List< [Alien](#) > [aliens](#))
Populates the aliens list with all alien objects from the game objects list.
- List< [Alien](#) > [getAllAliens](#) ([GamePanel](#) gamePanel)
Gets all alien objects from the game panel.
- void [handleAlienHit](#) ([PlayerProjectile](#) projectile, [GamePanel](#) gamePanel)
Checks for and handles collisions between a player projectile and aliens.
- boolean [hasAlienHitBottom](#) ([GamePanel](#) gamePanel, [Player](#) player)
Checks if any alien has reached the player's vertical position.
- boolean [areAllAliensDestroyed](#) ([GamePanel](#) gamePanel)
Checks if all aliens have been destroyed.

8.2.1 Detailed Description

Manages alien collective behavior including movement, collisions, and destruction.

8.2.2 Member Function Documentation

8.2.2.1 addAllAliens()

```
void en.tresz.spaceinvaders.game.AlienController.addAllAliens (
    List< GameObject > objects,
    List< Alien > aliens) [inline]
```

Populates the aliens list with all alien objects from the game objects list.

Parameters

<i>objects</i>	the list of all game objects
<i>aliens</i>	the list to populate with aliens

8.2.2.2 areAllAliensDestroyed()

```
boolean en.tresz.spaceinvaders.game.AlienController.areAllAliensDestroyed (
    GamePanel gamePanel) [inline]
```

Checks if all aliens have been destroyed.

Parameters

<i>gamePanel</i>	the game panel
------------------	----------------

Returns

true if no aliens remain

8.2.2.3 getAllAliens()

```
List< Alien > en.tresz.spaceinvaders.game.AlienController.getAllAliens (
    GamePanel gamePanel) [inline]
```

Gets all alien objects from the game panel.

Parameters

<i>gamePanel</i>	the game panel
------------------	----------------

Returns

a list of all aliens

8.2.2.4 handleAlienHit()

```
void en.tresz.spaceinvaders.game.AlienController.handleAlienHit (
    PlayerProjectile projectile,
    GamePanel gamePanel) [inline]
```

Checks for and handles collisions between a player projectile and aliens.

Parameters

<i>projectile</i>	the player projectile
<i>gamePanel</i>	the game panel

8.2.2.5 handleAllAlienCollisions()

```
void en.tresz.spaceinvaders.game.AlienController.handleAllAlienCollisions (
    GamePanel gamePanel) [inline]
```

Handles collisions between all aliens, reversing velocities and separating overlapping aliens.

Parameters

<i>gamePanel</i>	the game panel containing aliens
------------------	----------------------------------

8.2.2.6 hasAlienHitBottom()

```
boolean en.tresz.spaceinvaders.game.AlienController.hasAlienHitBottom (
    GamePanel gamePanel,
    Player player) [inline]
```

Checks if any alien has reached the player's vertical position.

Parameters

<i>gamePanel</i>	the game panel
<i>player</i>	the player

Returns

true if any alien reached the bottom

8.2.2.7 moveDownAllAliens()

```
void en.tresz.spaceinvaders.game.AlienController.moveDownAllAliens (
    GamePanel gamePanel) [inline]
```

Moves all aliens down by one row height.

Parameters

<i>gamePanel</i>	the game panel containing aliens
------------------	----------------------------------

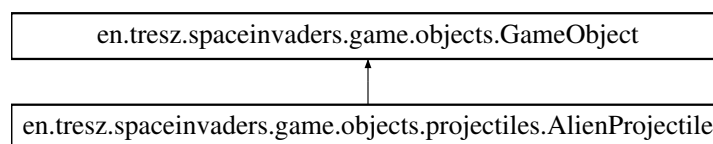
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/AlienController.java](#)

8.3 en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile Class Reference

A projectile fired by aliens moving downwards.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile:

**Public Member Functions**

- [AlienProjectile](#) ([Vector2D](#) position)
Constructs an [AlienProjectile](#) at the given position.
- void [update](#) ([GamePanel](#) gamePanel)
Updates the projectile position moving downwards.
- void [draw](#) (Graphics g)
Draws the projectile.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [Vector2D](#) [position](#)
- [Vector2D](#) [velocity](#)
- int [width](#)
- int [height](#)

8.3.1 Detailed Description

A projectile fired by aliens moving downwards.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 AlienProjectile()

```
en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile.AlienProjectile (
    Vector2D position) [inline]
```

Constructs an [AlienProjectile](#) at the given position.

Parameters

<i>position</i>	the starting position
-----------------	-----------------------

8.3.3 Member Function Documentation

8.3.3.1 draw()

```
void en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile.draw (
    Graphics g) [inline]
```

Draws the projectile.

Parameters

<i>g</i>	the graphics context
----------	----------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

8.3.3.2 update()

```
void en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile.update (
    GamePanel gamePanel) [inline]
```

Updates the projectile position moving downwards.

Parameters

<i>gamePanel</i>	the game panel (not used)
------------------	---------------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/objects/projectiles/AlienProjectile.java](#)

8.4 en.tresz.spaceinvaders.util.ButtonMaker Class Reference

A utility class to create customized buttons.

Static Public Member Functions

- static JButton [buttonSetup](#) (String name, int width, int height)

*Makes a customized button The names of the images should follow these rules: -normal icon: <name>-button.png
-hover icon: <name>-button-selected.gif -hover icon: <name>-button-pressed.gif.*

8.4.1 Detailed Description

A utility class to create customized buttons.

8.4.2 Member Function Documentation

8.4.2.1 `buttonSetup()`

```

JButton en.tresz.spaceinvaders.util.ButtonMaker.buttonSetup (
    String name,
    int width,
    int height) [inline], [static]

```

Makes a customized button The names of the images should follow these rules: -normal icon: <name>-button.png
-hover icon: <name>-button-selected.gif -hover icon: <name>-button-pressed.gif.

Parameters

<i>name</i>	the name of the button
<i>width</i>	the width of the button in pixels
<i>height</i>	the height of the button in pixels

Returns

a customized button

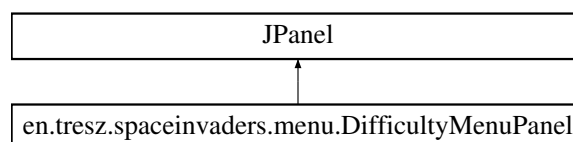
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/util/ButtonMaker.java](#)

8.5 `en.tresz.spaceinvaders.menu.DifficultyMenuPanel` Class Reference

The difficulty selection screen.

Inheritance diagram for `en.tresz.spaceinvaders.menu.DifficultyMenuPanel`:



Public Member Functions

- [DifficultyMenuPanel](#) ([MainWindow](#) gw)
Constructs a [DifficultyMenuPanel](#).

Protected Member Functions

- void [paintComponent](#) (Graphics g)
Paints the background image scaled to fit the panel.

8.5.1 Detailed Description

The difficulty selection screen.

Shows options for Easy, Medium, Hard, and a Back button.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 DifficultyMenuPanel()

```
en.tresz.spaceinvaders.menu.DifficultyMenuPanel.DifficultyMenuPanel (  
    MainWindow gw) [inline]
```

Constructs a [DifficultyMenuPanel](#).

Parameters

<i>gw</i>	the main game window
-----------	----------------------

8.5.3 Member Function Documentation

8.5.3.1 paintComponent()

```
void en.tresz.spaceinvaders.menu.DifficultyMenuPanel.paintComponent (  
    Graphics g) [inline], [protected]
```

Paints the background image scaled to fit the panel.

Parameters

<i>g</i>	the graphics context
----------	----------------------

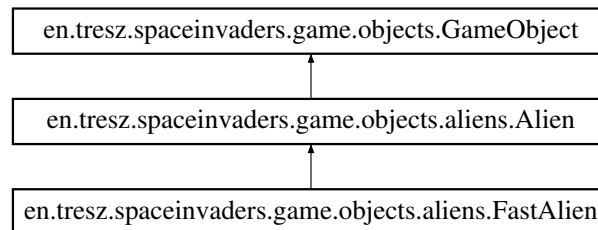
The documentation for this class was generated from the following file:

- src/main/java/en/tresz/spaceinvaders/menu/[DifficultyMenuPanel.java](#)

8.6 en.tresz.spaceinvaders.game.objects.aliens.FastAlien Class Reference

A fast-moving alien that moves three times faster than normal aliens.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.aliens.FastAlien:



Public Member Functions

- [FastAlien](#) ([Vector2D](#) p, [Vector2D](#) v, int hitSpeed)
Constructs a [FastAlien](#) with 3x horizontal velocity.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- void [update](#) ([GamePanel](#) gamePanel)
Updates the alien's position and handles wall collisions.
- void [shoot](#) ([GamePanel](#) gamePanel)
Attempts to shoot a projectile towards the player.
- void [draw](#) ([Graphics](#) g)
Draws the alien on the screen using its image.
- boolean [reachedBottom](#) ([Player](#) player)
Checks if the alien has reached the player's vertical position.
- boolean [isRequestGlobalMoveDown](#) ()
Checks if the alien requests all aliens to move down, resetting the flag after checking.
- int [getClosingInTreshold](#) ()
Gets the collision threshold before the alien requests a global move down.
- void [setCollisionCount](#) (int count)
Sets the collision count for this alien.
- int [getHitSpeed](#) ()
Gets the hit speed multiplier for this alien.
- boolean [chance](#) (double percentage)
Determines if a random event should occur based on a given percentage.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Static Public Attributes

- static final int [FAST_ALIEN_WIDTH](#) = 40

Static Public Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- static final int [ALIEN_HEIGHT](#) = 80

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- [Alien](#) ([Alien](#) alien)
Copy constructor for creating a new [Alien](#) from an existing one.
- [Alien](#) ([Vector2D](#) p, [Vector2D](#) v, int width, int hitSpeed)
Constructs a new [Alien](#) with the specified parameters.

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- BufferedImage [alienImage](#)
- int [collisionCount](#) = 0
- int [closingInTreshold](#) = 6
- int [hitSpeed](#)
- boolean [requestGlobalMoveDown](#) = false
- int [shootingInterval](#) = 500

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- Vector2D [position](#)
- Vector2D [velocity](#)
- int [width](#)
- int [height](#)

8.6.1 Detailed Description

A fast-moving alien that moves three times faster than normal aliens.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 FastAlien()

```
en.tresz.spaceinvaders.game.objects.aliens.FastAlien.FastAlien (
    Vector2D p,
    Vector2D v,
    int hitSpeed) [inline]
```

Constructs a [FastAlien](#) with 3x horizontal velocity.

Parameters

<i>p</i>	the initial position
<i>v</i>	the initial velocity (tripled horizontally)
<i>hitSpeed</i>	the speed multiplier

8.6.3 Member Data Documentation

8.6.3.1 FAST_ALIEN_WIDTH

```
final int en.tresz.spaceinvaders.game.objects.aliens.FastAlien.FAST_ALIEN_WIDTH = 40 [static]
```

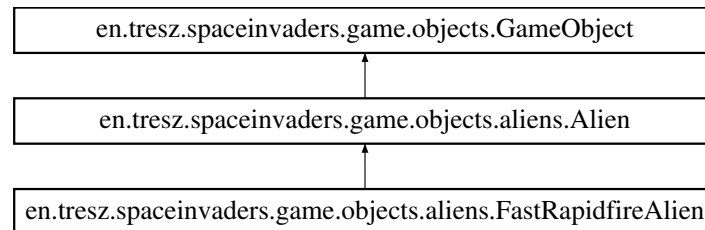
The documentation for this class was generated from the following file:

- src/main/java/en/tresz/spaceinvaders/game/objects/aliens/[FastAlien.java](#)

8.7 en.tresz.spaceinvaders.game.objects.aliens.FastRapidfireAlien Class Reference

A fast-moving alien that also shoots three times faster than normal.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.aliens.FastRapidfireAlien:



Public Member Functions

- [FastRapidfireAlien](#) ([Vector2D](#) p, [Vector2D](#) v, int hitSpeed)
Constructs a [FastRapidfireAlien](#) with 3x movement speed and 3x shooting speed.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- void [update](#) ([GamePanel](#) gamePanel)
Updates the alien's position and handles wall collisions.
- void [shoot](#) ([GamePanel](#) gamePanel)
Attempts to shoot a projectile towards the player.
- void [draw](#) ([Graphics](#) g)
Draws the alien on the screen using its image.
- boolean [reachedBottom](#) ([Player](#) player)
Checks if the alien has reached the player's vertical position.
- boolean [isRequestGlobalMoveDown](#) ()
Checks if the alien requests all aliens to move down, resetting the flag after checking.
- int [getClosingInThreshold](#) ()
Gets the collision threshold before the alien requests a global move down.
- void [setCollisionCount](#) (int count)
Sets the collision count for this alien.
- int [getHitSpeed](#) ()
Gets the hit speed multiplier for this alien.
- boolean [chance](#) (double percentage)
Determines if a random event should occur based on a given percentage.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Static Public Attributes

- static final int [FAST_RAPIDFIRE_ALIEN_WIDTH](#) = 40

Static Public Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- static final int [ALIEN_HEIGHT](#) = 80

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- [Alien](#) ([Alien](#) alien)
Copy constructor for creating a new [Alien](#) from an existing one.
- [Alien](#) ([Vector2D](#) p, [Vector2D](#) v, int width, int hitSpeed)
Constructs a new [Alien](#) with the specified parameters.

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- BufferedImage [alienImage](#)
- int [collisionCount](#) = 0
- int [closingInTreshold](#) = 6
- int [hitSpeed](#)
- boolean [requestGlobalMoveDown](#) = false
- int [shootingInterval](#) = 500

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- Vector2D [position](#)
- Vector2D [velocity](#)
- int [width](#)
- int [height](#)

8.7.1 Detailed Description

A fast-moving alien that also shoots three times faster than normal.

8.7.2 Constructor & Destructor Documentation**8.7.2.1 FastRapidfireAlien()**

```
en.tresz.spaceinvaders.game.objects.aliens.FastRapidfireAlien.FastRapidfireAlien (
    Vector2D p,
    Vector2D v,
    int hitSpeed) [inline]
```

Constructs a [FastRapidfireAlien](#) with 3x movement speed and 3x shooting speed.

Parameters

<i>p</i>	the initial position
<i>v</i>	the initial velocity (tripled horizontally)
<i>hitSpeed</i>	the base speed multiplier (tripled for shooting, doubled overall)

8.7.3 Member Data Documentation**8.7.3.1 FAST_RAPIDFIRE_ALIEN_WIDTH**

```
final int en.tresz.spaceinvaders.game.objects.aliens.FastRapidfireAlien.FAST_RAPIDFIRE_ALIEN←→
_WIDTH = 40 [static]
```

The documentation for this class was generated from the following file:

- src/main/java/en/tresz/spaceinvaders/game/objects/aliens/[FastRapidfireAlien.java](#)

8.8 en.tresz.spaceinvaders.game.objects.GameObject Class Reference

The base class for all game objects.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.GameObject:



Public Member Functions

- abstract void [update](#) ([GamePanel](#) gamePanel)
Updates the object state.
- abstract void [draw](#) ([Graphics](#) g)
Draws the object.
- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Protected Member Functions

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes

- [Vector2D](#) [position](#)
- [Vector2D](#) [velocity](#)
- int [width](#)
- int [height](#)

8.8.1 Detailed Description

The base class for all game objects.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 GameObject()

```
en.tresz.spaceinvaders.game.objects.GameObject.GameObject (
    Vector2D p,
    Vector2D v) [inline], [protected]
```

Constructor for the [GameObject](#) class.

Parameters

p	the position vector
v	the velocity vector

8.8.3 Member Function Documentation

8.8.3.1 draw()

```
abstract void en.tresz.spaceinvaders.game.objects.GameObject.draw (
    Graphics g) [abstract]
```

Draws the object.

Reimplemented in [en.tresz.spaceinvaders.game.objects.aliens.Alien](#), [en.tresz.spaceinvaders.game.objects.Player](#), [en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile](#), and [en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile](#).

8.8.3.2 getHalfHeight()

```
int en.tresz.spaceinvaders.game.objects.GameObject.getHalfHeight () [inline]
```

Gets half the height (used for collision detection, and drawing).

Returns

half the height

8.8.3.3 getHalfWidth()

```
int en.tresz.spaceinvaders.game.objects.GameObject.getHalfWidth () [inline]
```

Gets half the width (used for collision detection, and drawing).

Returns

half the width

8.8.3.4 getHeight()

```
int en.tresz.spaceinvaders.game.objects.GameObject.getHeight () [inline]
```

Gets the height of the object.

Returns

the height

8.8.3.5 getPosition()

```
Vector2D en.tresz.spaceinvaders.game.objects.GameObject.getPosition () [inline]
```

Gets the position vector.

Returns

the position

8.8.3.6 getVelocity()

```
Vector2D en.tresz.spaceinvaders.game.objects.GameObject.getVelocity () [inline]
```

Gets the velocity vector.

Returns

the velocity

8.8.3.7 getWidth()

```
int en.tresz.spaceinvaders.game.objects.GameObject.getWidth () [inline]
```

Gets the width of the object.

Returns

the width

8.8.3.8 intersects()

```
boolean en.tresz.spaceinvaders.game.objects.GameObject.intersects (  
    GameObject other) [inline]
```

Checks if this object intersects with another object.

Parameters

<i>other</i>	the other game object to check
--------------	--------------------------------

Returns

true if the objects intersect

8.8.3.9 setPosition()

```
void en.tresz.spaceinvaders.game.objects.GameObject.setPosition (  
    Vector2D p) [inline]
```

Sets the position vector.

Parameters

<i>p</i>	the new position
----------	------------------

8.8.3.10 setVelocity()

```
void en.tresz.spaceinvaders.game.objects.GameObject.setVelocity (  
    Vector2D v) [inline]
```

Sets the velocity vector.

Parameters

<i>v</i>	the new velocity
----------	------------------

8.8.3.11 update()

```
abstract void en.tresz.spaceinvaders.game.objects.GameObject.update (  
    GamePanel gamePanel) [abstract]
```

Updates the object state.

Parameters

<i>gamePanel</i>	the game panel
------------------	----------------

Reimplemented in [en.tresz.spaceinvaders.game.objects.aliens.Alien](#), [en.tresz.spaceinvaders.game.objects.Player](#), [en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile](#), and [en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile](#)

8.8.4 Member Data Documentation

8.8.4.1 height

`int en.tresz.spaceinvaders.game.objects.GameObject.height` [protected]

8.8.4.2 position

`Vector2D en.tresz.spaceinvaders.game.objects.GameObject.position` [protected]

8.8.4.3 velocity

`Vector2D en.tresz.spaceinvaders.game.objects.GameObject.velocity` [protected]

8.8.4.4 width

`int en.tresz.spaceinvaders.game.objects.GameObject.width` [protected]

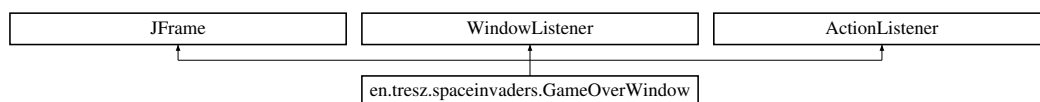
The documentation for this class was generated from the following file:

- `src/main/java/en/tresz/spaceinvaders/game/objects/GameObject.java`

8.9 en.tresz.spaceinvaders.GameOverWindow Class Reference

Game over window for easy/medium difficulty with play again option.

Inheritance diagram for `en.tresz.spaceinvaders.GameOverWindow`:



Public Member Functions

- `GameOverWindow` (int time, `MainWindow` mainWindow)
Constructs a `GameOverWindow` displaying the completion time.
- void `actionPerformed` (ActionEvent e)
Handles the play again button action.
- void `windowOpened` (WindowEvent e)
- void `windowClosing` (WindowEvent e)
- void `windowClosed` (WindowEvent e)
- void `windowIconified` (WindowEvent e)
- void `windowDeiconified` (WindowEvent e)
- void `windowActivated` (WindowEvent e)
- void `windowDeactivated` (WindowEvent e)

8.9.1 Detailed Description

Game over window for easy/medium difficulty with play again option.

8.9.2 Constructor & Destructor Documentation

8.9.2.1 GameOverWindow()

```
en.tresz.spaceinvaders.GameOverWindow.GameOverWindow (
    int time,
    MainWindow mainWindow) [inline]
```

Constructs a [GameOverWindow](#) displaying the completion time.

Parameters

<i>time</i>	the game completion time in seconds
<i>mainWindow</i>	the main window reference

8.9.3 Member Function Documentation

8.9.3.1 actionPerformed()

```
void en.tresz.spaceinvaders.GameOverWindow.actionPerformed (
    ActionEvent e) [inline]
```

Handles the play again button action.

Parameters

<i>e</i>	the action event
----------	------------------

8.9.3.2 windowActivated()

```
void en.tresz.spaceinvaders.GameOverWindow.windowActivated (
    WindowEvent e) [inline]
```

8.9.3.3 windowClosed()

```
void en.tresz.spaceinvaders.GameOverWindow.windowClosed (
    WindowEvent e) [inline]
```

8.9.3.4 windowClosing()

```
void en.tresz.spaceinvaders.GameOverWindow.windowClosing (
    WindowEvent e) [inline]
```

8.9.3.5 windowDeactivated()

```
void en.tresz.spaceinvaders.GameOverWindow.windowDeactivated (
    WindowEvent e) [inline]
```

8.9.3.6 windowDeiconified()

```
void en.tresz.spaceinvaders.GameOverWindow.windowDeiconified (
    WindowEvent e) [inline]
```

8.9.3.7 windowIconified()

```
void en.tresz.spaceinvaders.GameOverWindow.windowIconified (
    WindowEvent e) [inline]
```

8.9.3.8 windowOpened()

```
void en.tresz.spaceinvaders.GameOverWindow.windowOpened (
    WindowEvent e) [inline]
```

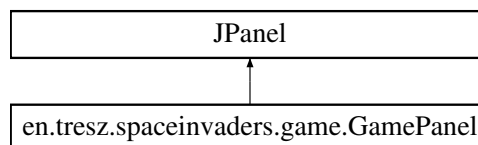
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/GameOverWindow.java](#)

8.10 en.tresz.spaceinvaders.game.GamePanel Class Reference

The panel where the game runs.

Inheritance diagram for en.tresz.spaceinvaders.game.GamePanel:



Public Member Functions

- [GamePanel](#) ([MainWindow](#) mw)
Constructs a [GamePanel](#).
- void [startGame](#) (String difficulty)
Starts the game with the given difficulty.
- boolean [updateGame](#) ()
Updates the game state including object movements, collisions, and shooting.
- void [createEasyObjects](#) ()
Creates game objects for easy difficulty.
- void [createMediumObjects](#) ()
Creates game objects for medium difficulty.
- void [createHardObjects](#) ()
Creates game objects for hard difficulty.
- List< [GameObject](#) > [getGameObjects](#) ()
Gets the list of game objects.
- List< [Alien](#) > [getAliens](#) ()
Gets the list of aliens.
- void [addGameObject](#) ([GameObject](#) object)
Adds a game object to the panel.
- void [removeGameObject](#) ([GameObject](#) object)
Removes a game object from the panel.
- [HealthBar](#) [getHealthBar](#) ()
Gets the health bar.
- [GameTimer](#) [getGameTimer](#) ()
Gets the game timer.
- void [setPlayerHit](#) (boolean playerHit)
Sets the player hit status.
- void [testAddObject](#) ([GameObject](#) object)
Test method to add a game object directly (for testing purposes).
- void [testRemoveObject](#) ([GameObject](#) object)
Test method to remove a game object directly (for testing purposes).

Protected Member Functions

- void [paintComponent](#) (Graphics g)
Paints the game objects.

8.10.1 Detailed Description

The panel where the game runs.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 [GamePanel](#)()

```
en.tresz.spaceinvaders.game.GamePanel.GamePanel (
    MainWindow mw) [inline]
```

Constructs a [GamePanel](#).

Parameters

<i>mw</i>	the main window reference
-----------	---------------------------

8.10.3 Member Function Documentation

8.10.3.1 addGameObject()

```
void en.tresz.spaceinvaders.game.GamePanel.addGameObject (
    GameObject object) [inline]
```

Adds a game object to the panel.

Parameters

<i>object</i>	the game object to add
---------------	------------------------

8.10.3.2 createEasyObjects()

```
void en.tresz.spaceinvaders.game.GamePanel.createEasyObjects () [inline]
```

Creates game objects for easy difficulty.

8.10.3.3 createHardObjects()

```
void en.tresz.spaceinvaders.game.GamePanel.createHardObjects () [inline]
```

Creates game objects for hard difficulty.

8.10.3.4 createMediumObjects()

```
void en.tresz.spaceinvaders.game.GamePanel.createMediumObjects () [inline]
```

Creates game objects for medium difficulty.

8.10.3.5 getAliens()

```
List< Alien > en.tresz.spaceinvaders.game.GamePanel.getAliens () [inline]
```

Gets the list of aliens.

Returns

the list of aliens

8.10.3.6 getGameObjects()

```
List< GameObject > en.tresz.spaceinvaders.game.GamePanel.getGameObjects () [inline]
```

Gets the list of game objects.

Returns

the list of game objects

8.10.3.7 getGameTimer()

```
GameTimer en.tresz.spaceinvaders.game.GamePanel.getGameTimer () [inline]
```

Gets the game timer.

Returns

the game timer

8.10.3.8 getHealthBar()

```
HealthBar en.tresz.spaceinvaders.game.GamePanel.getHealthBar () [inline]
```

Gets the health bar.

Returns

the health bar

8.10.3.9 paintComponent()

```
void en.tresz.spaceinvaders.game.GamePanel.paintComponent (  
    Graphics g) [inline], [protected]
```

Paints the game objects.

8.10.3.10 removeGameObject()

```
void en.tresz.spaceinvaders.game.GamePanel.removeGameObject (  
    GameObject object) [inline]
```

Removes a game object from the panel.

Parameters

<i>object</i>	the game object to remove
---------------	---------------------------

8.10.3.11 setPlayerHit()

```
void en.tresz.spaceinvaders.game.GamePanel.setPlayerHit (  
    boolean playerHit) [inline]
```

Sets the player hit status.

Parameters

<i>playerHit</i>	true if the player is hit, false otherwise
------------------	--

8.10.3.12 startGame()

```
void en.tresz.spaceinvaders.game.GamePanel.startGame (  
    String difficulty) [inline]
```

Starts the game with the given difficulty.

Parameters

<i>difficulty</i>	the difficulty of the game
-------------------	----------------------------

8.10.3.13 testAddObject()

```
void en.tresz.spaceinvaders.game.GamePanel.testAddObject (  
    GameObject object) [inline]
```

Test method to add a game object directly (for testing purposes).

!!!

Parameters

<i>object</i>	the game object to add
---------------	------------------------

8.10.3.14 testRemoveObject()

```
void en.tresz.spaceinvaders.game.GamePanel.testRemoveObject (  
    GameObject object) [inline]
```

Test method to remove a game object directly (for testing purposes).

!!!

Parameters

<i>object</i>	the game object to remove
---------------	---------------------------

8.10.3.15 updateGame()

```
boolean en.tresz.spaceinvaders.game.GamePanel.updateGame () [inline]
```

Updates the game state including object movements, collisions, and shooting.

Returns

false if the game is over, true otherwise

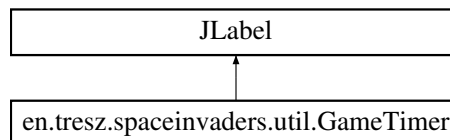
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/GamePanel.java](#)

8.11 en.tresz.spaceinvaders.util.GameTimer Class Reference

A label that displays and tracks game time in MM:SS format.

Inheritance diagram for en.tresz.spaceinvaders.util.GameTimer:



Public Member Functions

- [GameTimer](#) ()
Constructs a [GameTimer](#) and starts counting from 00:00.
- void [start](#) ()
Starts the timer.
- void [stop](#) ()
Stops the timer.
- void [reset](#) ()
Resets the timer to 00:00.
- int [getMinutes](#) ()
Gets the minutes component of the timer.
- int [getSeconds](#) ()
Gets the seconds component of the timer.
- [Time](#) [getTime](#) ()
Gets the [Time](#) object representing the current time.
- int [getTotalSeconds](#) ()
Gets the total time in seconds.

8.11.1 Detailed Description

A label that displays and tracks game time in MM:SS format.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 GameTimer()

```
en.tresz.spaceinvaders.util.GameTimer.GameTimer () [inline]
```

Constructs a [GameTimer](#) and starts counting from 00:00.

8.11.3 Member Function Documentation

8.11.3.1 getMinutes()

```
int en.tresz.spaceinvaders.util.GameTimer.getMinutes () [inline]
```

Gets the minutes component of the timer.

Returns

the minutes

8.11.3.2 getSeconds()

```
int en.tresz.spaceinvaders.util.GameTimer.getSeconds () [inline]
```

Gets the seconds component of the timer.

Returns

the seconds

8.11.3.3 getTime()

```
Time en.tresz.spaceinvaders.util.GameTimer.getTime () [inline]
```

Gets the [Time](#) object representing the current time.

Returns

the [Time](#) object

8.11.3.4 getTotalSeconds()

```
int en.tresz.spaceinvaders.util.GameTimer.getTotalSeconds () [inline]
```

Gets the total time in seconds.

Returns

the total seconds elapsed

8.11.3.5 reset()

```
void en.tresz.spaceinvaders.util.GameTimer.reset () [inline]
```

Resets the timer to 00:00.

8.11.3.6 start()

```
void en.tresz.spaceinvaders.util.GameTimer.start () [inline]
```

Starts the timer.

8.11.3.7 stop()

```
void en.tresz.spaceinvaders.util.GameTimer.stop () [inline]
```

Stops the timer.

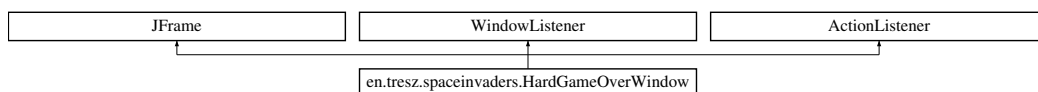
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/util/GameTimer.java](#)

8.12 en.tresz.spaceinvaders.HardGameOverWindow Class Reference

Game over window for hard difficulty that allows entering player name for scoreboard.

Inheritance diagram for en.tresz.spaceinvaders.HardGameOverWindow:



Public Member Functions

- [HardGameOverWindow](#) (int time, [MainWindow](#) mainWindow)
Constructs a [HardGameOverWindow](#) with time and score entry.
- String [getPlayerName](#) ()
Gets the player name entered.
- void [actionPerformed](#) (ActionEvent e)
Handles the enter button action to save the score.
- void [windowOpened](#) (WindowEvent e)
- void [windowClosing](#) (WindowEvent e)
- void [windowClosed](#) (WindowEvent e)
- void [windowIconified](#) (WindowEvent e)
- void [windowDeiconified](#) (WindowEvent e)
- void [windowActivated](#) (WindowEvent e)
- void [windowDeactivated](#) (WindowEvent e)

8.12.1 Detailed Description

Game over window for hard difficulty that allows entering player name for scoreboard.

8.12.2 Constructor & Destructor Documentation

8.12.2.1 HardGameOverWindow()

```
en.tresz.spaceinvaders.HardGameOverWindow.HardGameOverWindow (
    int time,
    MainWindow mainWindow) [inline]
```

Constructs a [HardGameOverWindow](#) with time and score entry.

Parameters

<i>time</i>	the game completion time in seconds
<i>mainWindow</i>	the main window reference

8.12.3 Member Function Documentation

8.12.3.1 actionPerformed()

```
void en.tresz.spaceinvaders.HardGameOverWindow.actionPerformed (
   (ActionEvent e) [inline]
```

Handles the enter button action to save the score.

Parameters

<i>e</i>	the action event
----------	------------------

8.12.3.2 getPlayerName()

```
String en.tresz.spaceinvaders.HardGameOverWindow.getPlayerName () [inline]
```

Gets the player name entered.

Returns

the player name

8.12.3.3 windowActivated()

```
void en.tresz.spaceinvaders.HardGameOverWindow.windowActivated (  
    WindowEvent e) [inline]
```

8.12.3.4 windowClosed()

```
void en.tresz.spaceinvaders.HardGameOverWindow.windowClosed (  
    WindowEvent e) [inline]
```

8.12.3.5 windowClosing()

```
void en.tresz.spaceinvaders.HardGameOverWindow.windowClosing (  
    WindowEvent e) [inline]
```

8.12.3.6 windowDeactivated()

```
void en.tresz.spaceinvaders.HardGameOverWindow.windowDeactivated (  
    WindowEvent e) [inline]
```

8.12.3.7 windowDeiconified()

```
void en.tresz.spaceinvaders.HardGameOverWindow.windowDeiconified (  
    WindowEvent e) [inline]
```

8.12.3.8 windowIconified()

```
void en.tresz.spaceinvaders.HardGameOverWindow.windowIconified (  
    WindowEvent e) [inline]
```

8.12.3.9 windowOpened()

```
void en.tresz.spaceinvaders.HardGameOverWindow.windowOpened (
    WindowEvent e) [inline]
```

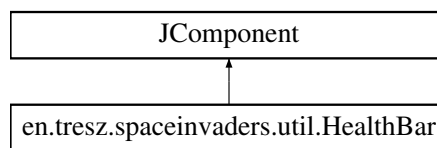
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/HardGameOverWindow.java](#)

8.13 en.tresz.spaceinvaders.util.HealthBar Class Reference

A visual health bar component that displays health as heart images.

Inheritance diagram for en.tresz.spaceinvaders.util.HealthBar:



Public Member Functions

- [HealthBar](#) (int maxHealth)
Constructs a [HealthBar](#) with the specified maximum health.
- void [takeDamage](#) (int amount)
Reduces the current health by the specified amount.
- void [reset](#) ()
Resets the health bar to maximum health.
- int [getCurrentHealth](#) ()
Gets the current health value.
- boolean [isDead](#) ()
Checks if the health has reached zero.

Protected Member Functions

- void [paintComponent](#) (Graphics g)
Paints the health bar by drawing heart images from right to left.

8.13.1 Detailed Description

A visual health bar component that displays health as heart images.

8.13.2 Constructor & Destructor Documentation

8.13.2.1 HealthBar()

```
en.tresz.spaceinvaders.util.HealthBar.HealthBar (
    int maxHealth) [inline]
```

Constructs a [HealthBar](#) with the specified maximum health.

Parameters

<i>maxHealth</i>	the maximum health value
------------------	--------------------------

8.13.3 Member Function Documentation

8.13.3.1 `getCurrentHealth()`

```
int en.tresz.spaceinvaders.util.HealthBar.getCurrentHealth () [inline]
```

Gets the current health value.

Returns

the current health

8.13.3.2 `isDead()`

```
boolean en.tresz.spaceinvaders.util.HealthBar.isDead () [inline]
```

Checks if the health has reached zero.

Returns

true if current health is zero or below

8.13.3.3 `paintComponent()`

```
void en.tresz.spaceinvaders.util.HealthBar.paintComponent (  
    Graphics g) [inline], [protected]
```

Paints the health bar by drawing heart images from right to left.

Parameters

<i>g</i>	the graphics context
----------	----------------------

8.13.3.4 `reset()`

```
void en.tresz.spaceinvaders.util.HealthBar.reset () [inline]
```

Resets the health bar to maximum health.

8.13.3.5 `takeDamage()`

```
void en.tresz.spaceinvaders.util.HealthBar.takeDamage (  
    int amount) [inline]
```

Reduces the current health by the specified amount.

Parameters

<i>amount</i>	the amount of damage to take
---------------	------------------------------

The documentation for this class was generated from the following file:

- `src/main/java/en/tresz/spaceinvaders/util/HealthBar.java`

8.14 en.tresz.spaceinvaders.util.ImageLoader Class Reference

A utility class to safely load images from the resources folder.

Static Public Member Functions

- static BufferedImage [loadBufferedImage](#) (String path)
Loads a BufferedImage from the resources folder.
- static BufferedImage [scaledBufferedImageNearest](#) (BufferedImage img, int width, int height)
Scales a BufferedImage to the given size using nearest-neighbor scaling.
- static ImageIcon [scaledIcon](#) (String path, int width, int height)
Loads a BufferedImage from the resources and returns a scaled ImageIcon using nearest-neighbor scaling.

8.14.1 Detailed Description

A utility class to safely load images from the resources folder.

8.14.2 Member Function Documentation

8.14.2.1 loadBufferedImage()

```
BufferedImage en.tresz.spaceinvaders.util.ImageLoader.loadBufferedImage (
    String path) [inline], [static]
```

Loads a BufferedImage from the resources folder.

Parameters

<i>path</i>	the resource path to the image
-------------	--------------------------------

Returns

the loaded BufferedImage, or null if loading fails

8.14.2.2 scaledBufferedImageNearest()

```
BufferedImage en.tresz.spaceinvaders.util.ImageLoader.scaledBufferedImageNearest (
    BufferedImage img,
    int width,
    int height) [inline], [static]
```

Scales a BufferedImage to the given size using nearest-neighbor scaling.

Parameters

<i>img</i>	the source image
<i>width</i>	the target width
<i>height</i>	the target height

Returns

the scaled image, or null if the source image is null

8.14.2.3 scaledIcon()

```
ImageIcon en.tresz.spaceinvaders.util.ImageLoader.scaledIcon (
    String path,
    int width,
    int height) [inline], [static]
```

Loads a BufferedImage from the resources and returns a scaled ImageIcon using nearest-neighbor scaling.

Parameters

<i>path</i>	resource path
<i>width</i>	target width in pixels
<i>height</i>	target height in pixels

Returns

scaled ImageIcon or null if resource missing

The documentation for this class was generated from the following file:

- src/main/java/en/tresz/spaceinvaders/util/[ImageLoader.java](#)

8.15 en.tresz.spaceinvaders.Main Class Reference

The entry point of the Space Invaders application.

Static Public Member Functions

- static void [main](#) (String[] args)

8.15.1 Detailed Description

The entry point of the Space Invaders application.

8.15.2 Member Function Documentation

8.15.2.1 main()

```
void en.tresz.spaceinvaders.Main.main (
    String[] args) [inline], [static]
```

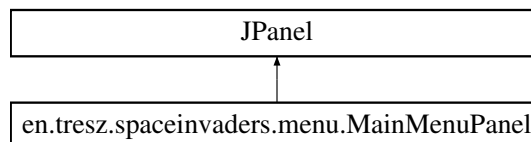
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/Main.java](#)

8.16 en.tresz.spaceinvaders.menu.MainMenuPanel Class Reference

The main menu panel with options to play, view scoreboard, or exit.

Inheritance diagram for en.tresz.spaceinvaders.menu.MainMenuPanel:



Public Member Functions

- [MainMenuPanel](#) ([MainWindow](#) gw)
Constructs a [MainMenuPanel](#).

Protected Member Functions

- void [paintComponent](#) (Graphics g)
Paints the background image scaled to fit the panel.

8.16.1 Detailed Description

The main menu panel with options to play, view scoreboard, or exit.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 MainMenuPanel()

```
en.tresz.spaceinvaders.menu.MainMenuPanel.MainMenuPanel (
    MainWindow gw) [inline]
```

Constructs a [MainMenuPanel](#).

Parameters

<i>gw</i>	the main game window
-----------	----------------------

8.16.3 Member Function Documentation

8.16.3.1 paintComponent()

```
void en.tresz.spaceinvaders.menu.MainMenuPanel.paintComponent (
    Graphics g) [inline], [protected]
```

Paints the background image scaled to fit the panel.

Parameters

<i>g</i>	the graphics context
----------	----------------------

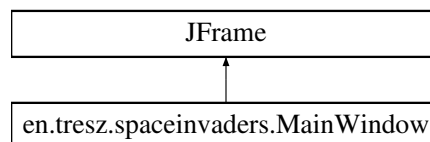
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/menu/MainMenuPanel.java](#)

8.17 en.tresz.spaceinvaders.MainWindow Class Reference

The main frame.

Inheritance diagram for en.tresz.spaceinvaders.MainWindow:



Classes

- enum [PanelType](#)

Public Member Functions

- [MainWindow](#) ()
Constructs the main window and initializes all panels.
- [GamePanel](#) [getGamePanel](#) ()
Gets the game panel.
- [ScoreboardPanel](#) [getScoreboardPanel](#) ()
Gets the scoreboard panel.
- [MainMenuPanel](#) [getMainMenuPanel](#) ()
Gets the main menu panel.
- [DifficultyMenuPanel](#) [getDifficultyMenuPanel](#) ()
Gets the difficulty menu panel.
- void [showPanel](#) ([PanelType](#) panelType)
Switches the current panel.
- void [startGame](#) (String difficulty)
Sets up the game and swaps to the game panel.

Static Public Attributes

- static final int `WINDOW_WIDTH` = 400
- static final int `WINDOW_HEIGHT` = 800

8.17.1 Detailed Description

The main frame.

8.17.2 Constructor & Destructor Documentation

8.17.2.1 MainWindow()

```
en.tresz.spaceinvaders.MainWindow.MainWindow () [inline]
```

Constructs the main window and initializes all panels.

8.17.3 Member Function Documentation

8.17.3.1 getDifficultyMenuPanel()

```
DifficultyMenuPanel en.tresz.spaceinvaders.MainWindow.getDifficultyMenuPanel () [inline]
```

Gets the difficulty menu panel.

Returns

the difficulty menu panel

8.17.3.2 getGamePanel()

```
GamePanel en.tresz.spaceinvaders.MainWindow.getGamePanel () [inline]
```

Gets the game panel.

Returns

the game panel

8.17.3.3 getMainMenuPanel()

```
MainMenuPanel en.tresz.spaceinvaders.MainWindow.getMainMenuPanel () [inline]
```

Gets the main menu panel.

Returns

the main menu panel

8.17.3.4 `getScoreboardPanel()`

```
ScoreboardPanel en.tresz.spaceinvaders.MainWindow.getScoreboardPanel () [inline]
```

Gets the scoreboard panel.

Returns

the scoreboard panel

8.17.3.5 `showPanel()`

```
void en.tresz.spaceinvaders.MainWindow.showPanel (  
    PanelType panelType) [inline]
```

Switches the current panel.

Parameters

<i>panelName</i>	the name of the panel to show
------------------	-------------------------------

8.17.3.6 `startGame()`

```
void en.tresz.spaceinvaders.MainWindow.startGame (  
    String difficulty) [inline]
```

Sets up the game and swaps to the game panel.

Parameters

<i>difficulty</i>	the difficulty of the game
-------------------	----------------------------

8.17.4 Member Data Documentation

8.17.4.1 `WINDOW_HEIGHT`

```
final int en.tresz.spaceinvaders.MainWindow.WINDOW_HEIGHT = 800 [static]
```

8.17.4.2 `WINDOW_WIDTH`

```
final int en.tresz.spaceinvaders.MainWindow.WINDOW_WIDTH = 400 [static]
```

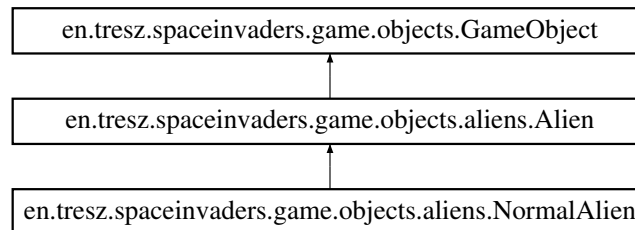
The documentation for this class was generated from the following file:

- `src/main/java/en/tresz/spaceinvaders/MainWindow.java`

8.18 en.tresz.spaceinvaders.game.objects.aliens.NormalAlien Class Reference

A standard alien with normal movement and shooting patterns.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.aliens.NormalAlien:



Public Member Functions

- [NormalAlien](#) ([Vector2D](#) p, [Vector2D](#) v, int hitSpeed)
Constructs a [NormalAlien](#).

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- void [update](#) ([GamePanel](#) gamePanel)
Updates the alien's position and handles wall collisions.
- void [shoot](#) ([GamePanel](#) gamePanel)
Attempts to shoot a projectile towards the player.
- void [draw](#) ([Graphics](#) g)
Draws the alien on the screen using its image.
- boolean [reachedBottom](#) ([Player](#) player)
Checks if the alien has reached the player's vertical position.
- boolean [isRequestGlobalMoveDown](#) ()
Checks if the alien requests all aliens to move down, resetting the flag after checking.
- int [getClosingInTreshold](#) ()
Gets the collision threshold before the alien requests a global move down.
- void [setCollisionCount](#) (int count)
Sets the collision count for this alien.
- int [getHitSpeed](#) ()
Gets the hit speed multiplier for this alien.
- boolean [chance](#) (double percentage)
Determines if a random event should occur based on a given percentage.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Static Public Attributes

- static final int [NORMAL_ALIEN_WIDTH](#) = 50

Static Public Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- static final int [ALIEN_HEIGHT](#) = 80

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- [Alien](#) ([Alien](#) alien)
Copy constructor for creating a new [Alien](#) from an existing one.
- [Alien](#) ([Vector2D](#) p, [Vector2D](#) v, int width, int hitSpeed)
Constructs a new [Alien](#) with the specified parameters.

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- BufferedImage [alienImage](#)
- int [collisionCount](#) = 0
- int [closingInTreshold](#) = 6
- int [hitSpeed](#)
- boolean [requestGlobalMoveDown](#) = false
- int [shootingInterval](#) = 500

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- Vector2D [position](#)
- Vector2D [velocity](#)
- int [width](#)
- int [height](#)

8.18.1 Detailed Description

A standard alien with normal movement and shooting patterns.

8.18.2 Constructor & Destructor Documentation

8.18.2.1 NormalAlien()

```
en.tresz.spaceinvaders.game.objects.aliens.NormalAlien.NormalAlien (
    Vector2D p,
    Vector2D v,
    int hitSpeed) [inline]
```

Constructs a [NormalAlien](#).

Parameters

<i>p</i>	the initial position
<i>v</i>	the initial velocity
<i>hitSpeed</i>	the speed multiplier

8.18.3 Member Data Documentation

8.18.3.1 NORMAL_ALIEN_WIDTH

```
final int en.tresz.spaceinvaders.game.objects.aliens.NormalAlien.NORMAL_ALIEN_WIDTH = 50 [static]
```

The documentation for this class was generated from the following file:

- src/main/java/en/tresz/spaceinvaders/game/objects/aliens/[NormalAlien.java](#)

8.19 en.tresz.spaceinvaders.MainWindow.PanelType Enum Reference

Public Attributes

- [MENU](#)
- [DIFFICULTY](#)
- [SCOREBOARD](#)
- [GAME](#)

8.19.1 Member Data Documentation

8.19.1.1 DIFFICULTY

`en.tresz.spaceinvaders.MainWindow.PanelType.DIFFICULTY`

8.19.1.2 GAME

`en.tresz.spaceinvaders.MainWindow.PanelType.GAME`

8.19.1.3 MENU

`en.tresz.spaceinvaders.MainWindow.PanelType.MENU`

8.19.1.4 SCOREBOARD

`en.tresz.spaceinvaders.MainWindow.PanelType.SCOREBOARD`

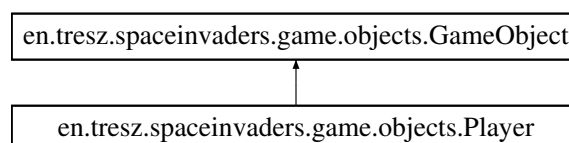
The documentation for this enum was generated from the following file:

- `src/main/java/en/tresz/spaceinvaders/MainWindow.java`

8.20 en.tresz.spaceinvaders.game.objects.Player Class Reference

Represents the player's spaceship controlled by keyboard input.

Inheritance diagram for `en.tresz.spaceinvaders.game.objects.Player`:



Public Member Functions

- [Player](#) ([Vector2D](#) position, [Vector2D](#) velocity, int hitSpeed, int health)
Constructs a [Player](#) with specified parameters.
- void [update](#) ([GamePanel](#) gamePanel)
Updates the player position and enforces boundary constraints.
- void [draw](#) ([Graphics](#) g)
Draws the player spaceship.
- void [draw](#) ([Graphics](#) g, boolean isHit)
Draws the player spaceship, showing hit state if applicable.
- void [shoot](#) ([GamePanel](#) gamePanel)
Attempts to shoot a projectile upwards.
- void [moveLeft](#) ()
Moves the player left.
- void [moveRight](#) ()
Moves the player right.
- void [stop](#) ()
Stops the player movement.
- void [playerMovement](#) ([GamePanel](#) gamePanel)
Sets up keyboard input bindings for player movement using arrow keys.
- int [getHealth](#) ()
Gets the player's current health.
- void [setHealth](#) (int health)
Sets the player's health.
- int [getHitSpeed](#) ()
Gets the player's hit speed.
- int [getShootingInterval](#) ()
Gets the player's shooting interval.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [Vector2D](#) position
- [Vector2D](#) velocity
- int width
- int height

8.20.1 Detailed Description

Represents the player's spaceship controlled by keyboard input.

8.20.2 Constructor & Destructor Documentation

8.20.2.1 Player()

```
en.tresz.spaceinvaders.game.objects.Player.Player (
    Vector2D position,
    Vector2D velocity,
    int hitSpeed,
    int health) [inline]
```

Constructs a [Player](#) with specified parameters.

Parameters

<i>position</i>	the initial position
<i>velocity</i>	the movement velocity
<i>hitSpeed</i>	the shooting speed multiplier
<i>health</i>	the initial health

8.20.3 Member Function Documentation

8.20.3.1 draw() [1/2]

```
void en.tresz.spaceinvaders.game.objects.Player.draw (
    Graphics g) [inline]
```

Draws the player spaceship.

Parameters

<i>g</i>	the graphics context
----------	----------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

8.20.3.2 draw() [2/2]

```
void en.tresz.spaceinvaders.game.objects.Player.draw (  
    Graphics g,  
    boolean isHit) [inline]
```

Draws the player spaceship, showing hit state if applicable.

Parameters

<i>g</i>	the graphics context
<i>isHit</i>	true if the player was hit, false otherwise

8.20.3.3 getHealth()

```
int en.tresz.spaceinvaders.game.objects.Player.getHealth () [inline]
```

Gets the player's current health.

Returns

the health

8.20.3.4 getHitSpeed()

```
int en.tresz.spaceinvaders.game.objects.Player.getHitSpeed () [inline]
```

Gets the player's hit speed.

Returns

the hit speed

8.20.3.5 getShootingInterval()

```
int en.tresz.spaceinvaders.game.objects.Player.getShootingInterval () [inline]
```

Gets the player's shooting interval.

Returns

the shooting interval in milliseconds

8.20.3.6 moveLeft()

```
void en.tresz.spaceinvaders.game.objects.Player.moveLeft () [inline]
```

Moves the player left.

8.20.3.7 moveRight()

```
void en.tresz.spaceinvaders.game.objects.Player.moveRight () [inline]
```

Moves the player right.

8.20.3.8 playerMovement()

```
void en.tresz.spaceinvaders.game.objects.Player.playerMovement (  
    GamePanel gamePanel) [inline]
```

Sets up keyboard input bindings for player movement using arrow keys.

Source: <https://www.javatips.net/api/javax.swing.inputmap>

Parameters

<i>gamePanel</i>	the game panel to register input bindings on
------------------	--

8.20.3.9 setHealth()

```
void en.tresz.spaceinvaders.game.objects.Player.setHealth (  
    int health) [inline]
```

Sets the player's health.

Parameters

<i>health</i>	the new health value
---------------	----------------------

8.20.3.10 shoot()

```
void en.tresz.spaceinvaders.game.objects.Player.shoot (  
    GamePanel gamePanel) [inline]
```

Attempts to shoot a projectile upwards.

Parameters

<i>gamePanel</i>	the game panel to add the projectile to
------------------	---

8.20.3.11 stop()

```
void en.tresz.spaceinvaders.game.objects.Player.stop () [inline]
```

Stops the player movement.

8.20.3.12 update()

```
void en.tresz.spaceinvaders.game.objects.Player.update (
    GamePanel gamePanel) [inline]
```

Updates the player position and enforces boundary constraints.

Parameters

<i>gamePanel</i>	the game panel for boundary checking
------------------	--------------------------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/objects/Player.java](#)

8.21 en.tresz.spaceinvaders.game.PlayerController Class Reference

Handles player-related game logic including collision detection with alien projectiles.

Public Member Functions

- void [handlePlayerHit](#) ([Player](#) player, [GamePanel](#) gamePanel)
Checks for and handles collisions between the player and alien projectiles.

8.21.1 Detailed Description

Handles player-related game logic including collision detection with alien projectiles.

8.21.2 Member Function Documentation

8.21.2.1 handlePlayerHit()

```
void en.tresz.spaceinvaders.game.PlayerController.handlePlayerHit (
    Player player,
    GamePanel gamePanel) [inline]
```

Checks for and handles collisions between the player and alien projectiles.

Parameters

<i>player</i>	the player to check
<i>gamePanel</i>	the game panel containing game objects

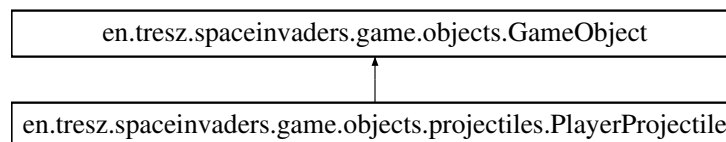
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/PlayerController.java](#)

8.22 en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile Class Reference

A projectile fired by the player moving upwards.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile:



Public Member Functions

- [PlayerProjectile](#) ([Vector2D](#) position)
Constructs a [PlayerProjectile](#) at the given position.
- void [update](#) ([GamePanel](#) gamePanel)
Updates the projectile position moving upwards.
- void [draw](#) ([Graphics](#) g)
Draws the projectile.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)

Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [Vector2D](#) position
- [Vector2D](#) velocity
- int width
- int height

8.22.1 Detailed Description

A projectile fired by the player moving upwards.

8.22.2 Constructor & Destructor Documentation

8.22.2.1 PlayerProjectile()

```
en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile.PlayerProjectile (
    Vector2D position) [inline]
```

Constructs a [PlayerProjectile](#) at the given position.

Parameters

<i>position</i>	the starting position
-----------------	-----------------------

8.22.3 Member Function Documentation

8.22.3.1 draw()

```
void en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile.draw (
    Graphics g) [inline]
```

Draws the projectile.

Parameters

<i>g</i>	the graphics context
----------	----------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

8.22.3.2 update()

```
void en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile.update (  
    GamePanel gamePanel) [inline]
```

Updates the projectile position moving upwards.

Parameters

<code>gamePanel</code>	the game panel (not used)
------------------------	---------------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.GameObject](#).

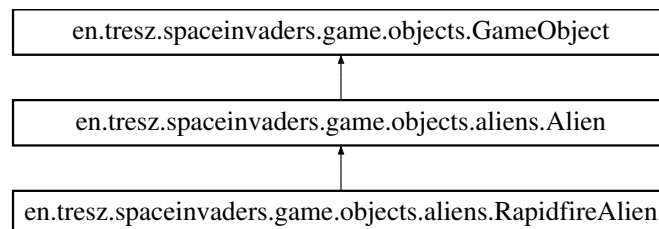
The documentation for this class was generated from the following file:

- `src/main/java/en/tresz/spaceinvaders/game/objects/projectiles/PlayerProjectile.java`

8.23 en.tresz.spaceinvaders.game.objects.aliens.RapidfireAlien Class Reference

An alien that shoots three times faster than normal aliens.

Inheritance diagram for `en.tresz.spaceinvaders.game.objects.aliens.RapidfireAlien`:



Public Member Functions

- [RapidfireAlien](#) ([Vector2D](#) p, [Vector2D](#) v, int hitSpeed)
Constructs a [RapidfireAlien](#) with 3x shooting speed.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- void [update](#) ([GamePanel](#) gamePanel)
Updates the alien's position and handles wall collisions.
- void [shoot](#) ([GamePanel](#) gamePanel)
Attempts to shoot a projectile towards the player.
- void [draw](#) ([Graphics](#) g)
Draws the alien on the screen using its image.
- boolean [reachedBottom](#) ([Player](#) player)
Checks if the alien has reached the player's vertical position.
- boolean [isRequestGlobalMoveDown](#) ()
Checks if the alien requests all aliens to move down, resetting the flag after checking.
- int [getClosingInThreshold](#) ()
Gets the collision threshold before the alien requests a global move down.
- void [setCollisionCount](#) (int count)
Sets the collision count for this alien.
- int [getHitSpeed](#) ()
Gets the hit speed multiplier for this alien.
- boolean [chance](#) (double percentage)
Determines if a random event should occur based on a given percentage.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Static Public Attributes

- static final int [RAPIDFIRE_ALIEN_WIDTH](#) = 60

Static Public Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- static final int [ALIEN_HEIGHT](#) = 80

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- [Alien](#) ([Alien](#) alien)
Copy constructor for creating a new [Alien](#) from an existing one.
- [Alien](#) ([Vector2D](#) p, [Vector2D](#) v, int width, int hitSpeed)
Constructs a new [Alien](#) with the specified parameters.

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) ([Vector2D](#) p, [Vector2D](#) v)
Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- BufferedImage [alienImage](#)
- int [collisionCount](#) = 0
- int [closingInTreshold](#) = 6
- int [hitSpeed](#)
- boolean [requestGlobalMoveDown](#) = false
- int [shootingInterval](#) = 500

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [Vector2D](#) [position](#)
- [Vector2D](#) [velocity](#)
- int [width](#)
- int [height](#)

8.23.1 Detailed Description

An alien that shoots three times faster than normal aliens.

8.23.2 Constructor & Destructor Documentation

8.23.2.1 RapidfireAlien()

```
en.tresz.spaceinvaders.game.objects.aliens.RapidfireAlien.RapidfireAlien (
    Vector2D p,
    Vector2D v,
    int hitSpeed) [inline]
```

Constructs a [RapidfireAlien](#) with 3x shooting speed.

Parameters

<i>p</i>	the initial position
<i>v</i>	the initial velocity
<i>hitSpeed</i>	the base speed multiplier (tripled for shooting)

8.23.3 Member Data Documentation

8.23.3.1 RAPIDFIRE_ALIEN_WIDTH

```
final int en.tresz.spaceinvaders.game.objects.aliens.RapidfireAlien.RAPIDFIRE_ALIEN_WIDTH = 60
[static]
```

The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/objects/aliens/RapidfireAlien.java](#)

8.24 en.tresz.spaceinvaders.util.Score Class Reference

Represents a player's score with their name and completion time.

Public Member Functions

- [Score](#) (String playerName, [Time](#) time)
Constructs a [Score](#) with player name and time.
- String [getPlayerName](#) ()
Gets the player's name.
- [Time](#) [getTime](#) ()
Gets the time achieved.

8.24.1 Detailed Description

Represents a player's score with their name and completion time.

8.24.2 Constructor & Destructor Documentation

8.24.2.1 Score()

```
en.tresz.spaceinvaders.util.Score.Score (
    String playerName,
    Time time) [inline]
```

Constructs a [Score](#) with player name and time.

Parameters

<i>playerName</i>	the name of the player
<i>time</i>	the time achieved

8.24.3 Member Function Documentation

8.24.3.1 getPlayerName()

```
String en.tresz.spaceinvaders.util.Score.getPlayerName () [inline]
```

Gets the player's name.

Returns

the player name

8.24.3.2 getTime()

```
Time en.tresz.spaceinvaders.util.Score.getTime () [inline]
```

Gets the time achieved.

Returns

the time

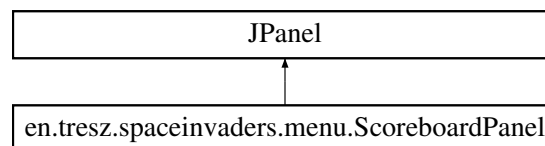
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/util/Score.java](#)

8.25 en.tresz.spaceinvaders.menu.ScoreboardPanel Class Reference

The scoreboard panel displays the top 5 high scores.

Inheritance diagram for en.tresz.spaceinvaders.menu.ScoreboardPanel:



Public Member Functions

- [ScoreboardPanel \(MainWindow gw\)](#)
Constructs a [ScoreboardPanel](#).
- void [refreshScores \(\)](#)
Refreshes the displayed scores by reloading from the JSON file.

8.25.1 Detailed Description

The scoreboard panel displays the top 5 high scores.

8.25.2 Constructor & Destructor Documentation

8.25.2.1 ScoreboardPanel()

```
en.tresz.spaceinvaders.menu.ScoreboardPanel.ScoreboardPanel (
    MainWindow gw) [inline]
```

Constructs a [ScoreboardPanel](#).

Parameters

<i>gw</i>	the main game window
-----------	----------------------

8.25.3 Member Function Documentation

8.25.3.1 refreshScores()

```
void en.tresz.spaceinvaders.menu.ScoreboardPanel.refreshScores () [inline]
```

Refreshes the displayed scores by reloading from the JSON file.

The documentation for this class was generated from the following file:

- src/main/java/en/tresz/spaceinvaders/menu/[ScoreboardPanel.java](#)

8.26 en.tresz.spaceinvaders.util.ScoreManager Class Reference

Manages saving and loading of high scores using JSON serialization.

Public Member Functions

- [ScoreManager](#) (int maxPlacements)
Constructs a [ScoreManager](#) with a custom maximum number of placements.
- [ScoreManager](#) ()
Constructs a [ScoreManager](#) with default maximum placements (5).
- void [addScore](#) ([Score](#) score)
Adds a score to the leaderboard in order.
- List< [Score](#) > [loadScores](#) ()
Loads scores from the JSON file.
- void [orderScores](#) (List< [Score](#) > scores)
Orders scores by time in ascending order (fastest first).

8.26.1 Detailed Description

Manages saving and loading of high scores using JSON serialization.

8.26.2 Constructor & Destructor Documentation

8.26.2.1 ScoreManager() [1/2]

```
en.tresz.spaceinvaders.util.ScoreManager.ScoreManager (  
    int maxPlacements) [inline]
```

Constructs a [ScoreManager](#) with a custom maximum number of placements.

Parameters

<i>maxPlacements</i>	the maximum number of scores to keep
----------------------	--------------------------------------

8.26.2.2 ScoreManager() [2/2]

```
en.tresz.spaceinvaders.util.ScoreManager.ScoreManager () [inline]
```

Constructs a [ScoreManager](#) with default maximum placements (5).

8.26.3 Member Function Documentation

8.26.3.1 addScore()

```
void en.tresz.spaceinvaders.util.ScoreManager.addScore (
    Score score) [inline]
```

Adds a score to the leaderboard in order.

Updates existing player scores if the new time is better.

Parameters

<i>score</i>	the score to add
--------------	------------------

8.26.3.2 loadScores()

```
List< Score > en.tresz.spaceinvaders.util.ScoreManager.loadScores () [inline]
```

Loads scores from the JSON file.

Returns

a list of scores, or an empty list if the file doesn't exist

8.26.3.3 orderScores()

```
void en.tresz.spaceinvaders.util.ScoreManager.orderScores (
    List< Score > scores) [inline]
```

Orders scores by time in ascending order (fastest first).

Parameters

<i>scores</i>	the list of scores to order
---------------	-----------------------------

The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/util/ScoreManager.java](#)

8.27 en.tresz.spaceinvaders.util.Time Class Reference

Represents a time duration in minutes and seconds.

Public Member Functions

- [Time](#) (int minutes, int seconds)
Constructs a [Time](#) object with the specified minutes and seconds.
- int [getTimeInSeconds](#) ()
Converts the time to total seconds.
- int [getMinutes](#) ()
Gets the minutes component.
- int [getSeconds](#) ()
Gets the seconds component.
- void [setMinutes](#) (int minutes)
Sets the minutes component.
- void [setSeconds](#) (int seconds)
Sets the seconds component.

8.27.1 Detailed Description

Represents a time duration in minutes and seconds.

8.27.2 Constructor & Destructor Documentation

8.27.2.1 Time()

```
en.tresz.spaceinvaders.util.Time.Time (  
    int minutes,  
    int seconds) [inline]
```

Constructs a [Time](#) object with the specified minutes and seconds.

Parameters

<i>minutes</i>	the minutes component
<i>seconds</i>	the seconds component

8.27.3 Member Function Documentation

8.27.3.1 getMinutes()

```
int en.tresz.spaceinvaders.util.Time.getMinutes () [inline]
```

Gets the minutes component.

Returns

the minutes

8.27.3.2 getSeconds()

```
int en.tresz.spaceinvaders.util.Time.getSeconds () [inline]
```

Gets the seconds component.

Returns

the seconds

8.27.3.3 getTimeInSeconds()

```
int en.tresz.spaceinvaders.util.Time.getTimeInSeconds () [inline]
```

Converts the time to total seconds.

Returns

the total time in seconds

8.27.3.4 setMinutes()

```
void en.tresz.spaceinvaders.util.Time.setMinutes (  
    int minutes) [inline]
```

Sets the minutes component.

Parameters

<i>minutes</i>	the new minutes value
----------------	-----------------------

8.27.3.5 setSeconds()

```
void en.tresz.spaceinvaders.util.Time.setSeconds (  
    int seconds) [inline]
```

Sets the seconds component.

Parameters

<i>seconds</i>	the new seconds value
----------------	-----------------------

The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/util/Time.java](#)

8.28 en.tresz.spaceinvaders.util.Vector2D Class Reference

2D vector class for positioning.

Public Member Functions

- [Vector2D](#) (int x, int y)
Constructor for the [Vector2D](#) class.
- int [getX](#) ()
Gets the x coordinate.
- int [getY](#) ()
Gets the y coordinate.
- void [setX](#) (int x)
Sets the x coordinate.
- void [setY](#) (int y)
Sets the y coordinate.
- int [centerX](#) (int width)
Centers the x coordinate based on the given width.
- int [centerY](#) (int height)
Centers the y coordinate based on the given height.
- void [add](#) ([Vector2D](#) v)
Adds another vector to this vector.
- void [addX](#) (int x)
Adds to the x coordinate.
- void [addY](#) (int y)
Adds to the y coordinate.
- void [subtract](#) ([Vector2D](#) v)
Subtracts another vector from this vector.
- void [subtractX](#) (int x)
Subtracts from the x coordinate.
- void [subtractY](#) (int y)
Subtracts from the y coordinate.
- void [negate](#) ()
Negates both coordinates.
- void [negateX](#) ()
Negates the x coordinate.
- void [negateY](#) ()
Negates the y coordinate.

8.28.1 Detailed Description

2D vector class for positioning.

8.28.2 Constructor & Destructor Documentation

8.28.2.1 Vector2D()

```
en.tresz.spaceinvaders.util.Vector2D.Vector2D (
    int x,
    int y) [inline]
```

Constructor for the [Vector2D](#) class.

Parameters

<i>x</i>	the x coordinate
<i>y</i>	the y coordinate

8.28.3 Member Function Documentation

8.28.3.1 add()

```
void en.tresz.spaceinvaders.util.Vector2D.add (  
    Vector2D v) [inline]
```

Adds another vector to this vector.

Parameters

<i>v</i>	the vector to add
----------	-------------------

8.28.3.2 addX()

```
void en.tresz.spaceinvaders.util.Vector2D.addX (  
    int x) [inline]
```

Adds to the x coordinate.

Parameters

<i>x</i>	the value to add to x
----------	-----------------------

8.28.3.3 addY()

```
void en.tresz.spaceinvaders.util.Vector2D.addY (  
    int y) [inline]
```

Adds to the y coordinate.

Parameters

<i>y</i>	the value to add to y
----------	-----------------------

8.28.3.4 centerX()

```
int en.tresz.spaceinvaders.util.Vector2D.centerX (  
    int width) [inline]
```

Centers the x coordinate based on the given width.

Parameters

<i>width</i>	the width to center around
--------------	----------------------------

Returns

the centered x coordinate

8.28.3.5 centerY()

```
int en.tresz.spaceinvaders.util.Vector2D.centerY (  
    int height) [inline]
```

Centers the y coordinate based on the given height.

Parameters

<i>height</i>	the height to center around
---------------	-----------------------------

Returns

the centered y coordinate

8.28.3.6 getX()

```
int en.tresz.spaceinvaders.util.Vector2D.getX () [inline]
```

Gets the x coordinate.

Returns

the x coordinate

8.28.3.7 getY()

```
int en.tresz.spaceinvaders.util.Vector2D.getY () [inline]
```

Gets the y coordinate.

Returns

the y coordinate

8.28.3.8 negate()

```
void en.tresz.spaceinvaders.util.Vector2D.negate () [inline]
```

Negates both coordinates.

8.28.3.9 negateX()

```
void en.tresz.spaceinvaders.util.Vector2D.negateX () [inline]
```

Negates the x coordinate.

8.28.3.10 negateY()

```
void en.tresz.spaceinvaders.util.Vector2D.negateY () [inline]
```

Negates the y coordinate.

8.28.3.11 setX()

```
void en.tresz.spaceinvaders.util.Vector2D.setX (  
    int x) [inline]
```

Sets the x coordinate.

Parameters

<i>x</i>	the new x coordinate
----------	----------------------

8.28.3.12 setY()

```
void en.tresz.spaceinvaders.util.Vector2D.setY (  
    int y) [inline]
```

Sets the y coordinate.

Parameters

<i>y</i>	the new y coordinate
----------	----------------------

8.28.3.13 subtract()

```
void en.tresz.spaceinvaders.util.Vector2D.subtract (  
    Vector2D v) [inline]
```

Subtracts another vector from this vector.

Parameters

<i>v</i>	the vector to subtract
----------	------------------------

8.28.3.14 subtractX()

```
void en.tresz.spaceinvaders.util.Vector2D.subtractX (  
    int x) [inline]
```

Subtracts from the x coordinate.

Parameters

x	the value to subtract from x
---	------------------------------

8.28.3.15 subtractY()

```
void en.tresz.spaceinvaders.util.Vector2D.subtractY (
    int y) [inline]
```

Subtracts from the y coordinate.

Parameters

y	the value to subtract from y
---	------------------------------

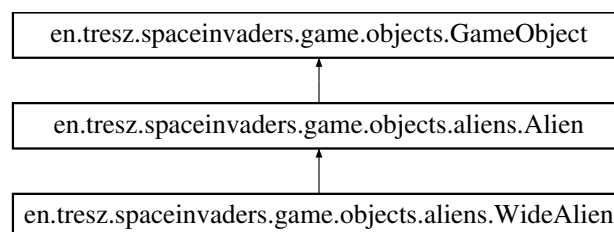
The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/util/Vector2D.java](#)

8.29 en.tresz.spaceinvaders.game.objects.aliens.WideAlien Class Reference

A wide alien that shoots three projectiles simultaneously.

Inheritance diagram for en.tresz.spaceinvaders.game.objects.aliens.WideAlien:



Public Member Functions

- [WideAlien](#) ([Vector2D](#) p, [Vector2D](#) v, int [hitSpeed](#))
Constructs a [WideAlien](#).
- void [shoot](#) ([GamePanel](#) gamePanel)
Shoots three projectiles: center, left, and right.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- void [update](#) ([GamePanel](#) gamePanel)
Updates the alien's position and handles wall collisions.
- void [draw](#) ([Graphics](#) g)
Draws the alien on the screen using its image.
- boolean [reachedBottom](#) ([Player](#) player)
Checks if the alien has reached the player's vertical position.
- boolean [isRequestGlobalMoveDown](#) ()
Checks if the alien requests all aliens to move down, resetting the flag after checking.
- int [getClosingInTreshold](#) ()
Gets the collision threshold before the alien requests a global move down.
- void [setCollisionCount](#) (int count)
Sets the collision count for this alien.
- int [getHitSpeed](#) ()
Gets the hit speed multiplier for this alien.
- boolean [chance](#) (double percentage)
Determines if a random event should occur based on a given percentage.

Public Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- boolean [intersects](#) ([GameObject](#) other)
Checks if this object intersects with another object.
- [Vector2D](#) [getPosition](#) ()
Gets the position vector.
- [Vector2D](#) [getVelocity](#) ()
Gets the velocity vector.
- void [setPosition](#) ([Vector2D](#) p)
Sets the position vector.
- void [setVelocity](#) ([Vector2D](#) v)
Sets the velocity vector.
- int [getWidth](#) ()
Gets the width of the object.
- int [getHeight](#) ()
Gets the height of the object.
- int [getHalfWidth](#) ()
Gets half the width (used for collision detection, and drawing).
- int [getHalfHeight](#) ()
Gets half the height (used for collision detection, and drawing).

Static Public Attributes

- static final int [WIDE_ALIEN_WIDTH](#) = 65

Static Public Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- static final int [ALIEN_HEIGHT](#) = 80

Additional Inherited Members

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- [Alien](#) (Alien alien)
Copy constructor for creating a new [Alien](#) from an existing one.
- [Alien](#) (Vector2D p, Vector2D v, int width, int hitSpeed)
Constructs a new [Alien](#) with the specified parameters.

Protected Member Functions inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- [GameObject](#) (Vector2D p, Vector2D v)
Constructor for the [GameObject](#) class.

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)

- BufferedImage [alienImage](#)
- int [collisionCount](#) = 0
- int [closingInThreshold](#) = 6
- int [hitSpeed](#)
- boolean [requestGlobalMoveDown](#) = false
- int [shootingInterval](#) = 500

Protected Attributes inherited from [en.tresz.spaceinvaders.game.objects.GameObject](#)

- Vector2D [position](#)
- Vector2D [velocity](#)
- int [width](#)
- int [height](#)

8.29.1 Detailed Description

A wide alien that shoots three projectiles simultaneously.

8.29.2 Constructor & Destructor Documentation

8.29.2.1 WideAlien()

```
en.tresz.spaceinvaders.game.objects.aliens.WideAlien.WideAlien (
    Vector2D p,
    Vector2D v,
    int hitSpeed) [inline]
```

Constructs a [WideAlien](#).

Parameters

<i>p</i>	the initial position
<i>v</i>	the initial velocity
<i>hitSpeed</i>	the speed multiplier

8.29.3 Member Function Documentation

8.29.3.1 shoot()

```
void en.tresz.spaceinvaders.game.objects.aliens.WideAlien.shoot (  
    GamePanel gamePanel) [inline]
```

Shoots three projectiles: center, left, and right.

Parameters

<i>gamePanel</i>	the game panel to add projectiles to
------------------	--------------------------------------

Reimplemented from [en.tresz.spaceinvaders.game.objects.aliens.Alien](#).

8.29.4 Member Data Documentation

8.29.4.1 WIDE_ALIEN_WIDTH

```
final int en.tresz.spaceinvaders.game.objects.aliens.WideAlien.WIDE_ALIEN_WIDTH = 65 [static]
```

The documentation for this class was generated from the following file:

- [src/main/java/en/tresz/spaceinvaders/game/objects/aliens/WideAlien.java](#)

Chapter 9

File Documentation

9.1 src/main/java/en/tresz/spaceinvaders/game/AlienController.java File Reference

```
import en.tresz.spaceinvaders.game.objects.GameObject;
import en.tresz.spaceinvaders.game.objects.Player;
import en.tresz.spaceinvaders.game.objects.aliens.Alien;
import en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile;
import java.util.List;
import java.util.ArrayList;
```

Classes

- class [en.tresz.spaceinvaders.game.AlienController](#)
Manages alien collective behavior including movement, collisions, and destruction.

Packages

- package [en.tresz.spaceinvaders.game](#)

9.2 src/main/java/en/tresz/spaceinvaders/game/GamePanel.java File Reference

```
import en.tresz.spaceinvaders.GameOverWindow;
import en.tresz.spaceinvaders.HardGameOverWindow;
import en.tresz.spaceinvaders.MainWindow;
import en.tresz.spaceinvaders.game.objects.GameObject;
import en.tresz.spaceinvaders.game.objects.Player;
import en.tresz.spaceinvaders.game.objects.aliens.*;
import en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile;
import en.tresz.spaceinvaders.util.*;
import java.util.ArrayList;
import java.util.List;
import javax.swing.JPanel;
import javax.swing.Timer;
import java.awt.Graphics;
import java.awt.image.BufferedImage;
```

Classes

- class [en.tresz.spaceinvaders.game.GamePanel](#)
The panel where the game runs.

Packages

- package [en.tresz.spaceinvaders.game](#)

9.3 src/main/java/en/tresz/spaceinvaders/game/objects/aliens/Alien.java File Reference

```
import en.tresz.spaceinvaders.game.GamePanel;
import en.tresz.spaceinvaders.game.objects.GameObject;
import en.tresz.spaceinvaders.game.objects.Player;
import en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile;
import en.tresz.spaceinvaders.util.*;
import java.awt.Graphics;
import java.awt.image.BufferedImage;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.aliens.Alien](#)
The [Alien](#) class represents a generic alien.

Packages

- package [en.tresz.spaceinvaders.game.objects.aliens](#)

9.4 src/main/java/en/tresz/spaceinvaders/game/objects/aliens/FastAlien.java File Reference

```
import en.tresz.spaceinvaders.util.ImageLoader;
import en.tresz.spaceinvaders.util.Vector2D;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.aliens.FastAlien](#)
A fast-moving alien that moves three times faster than normal aliens.

Packages

- package [en.tresz.spaceinvaders.game.objects.aliens](#)

9.5 src/main/java/en/tresz/spaceinvaders/game/objects/aliens/FastRapidfireAlien.java File Reference

```
import en.tresz.spaceinvaders.util.ImageLoader;
import en.tresz.spaceinvaders.util.Vector2D;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.aliens.FastRapidfireAlien](#)
A fast-moving alien that also shoots three times faster than normal.

Packages

- package [en.tresz.spaceinvaders.game.objects.aliens](#)

9.6 src/main/java/en/tresz/spaceinvaders/game/objects/aliens/NormalAlien.java File Reference

```
import en.tresz.spaceinvaders.util.ImageLoader;
import en.tresz.spaceinvaders.util.Vector2D;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.aliens.NormalAlien](#)
A standard alien with normal movement and shooting patterns.

Packages

- package [en.tresz.spaceinvaders.game.objects.aliens](#)

9.7 src/main/java/en/tresz/spaceinvaders/game/objects/aliens/RapidfireAlien.java File Reference

```
import en.tresz.spaceinvaders.util.ImageLoader;
import en.tresz.spaceinvaders.util.Vector2D;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.aliens.RapidfireAlien](#)
An alien that shoots three times faster than normal aliens.

Packages

- package [en.tresz.spaceinvaders.game.objects.aliens](#)

9.8 src/main/java/en/tresz/spaceinvaders/game/objects/aliens/WideAlien.java File Reference

```
import en.tresz.spaceinvaders.game.GamePanel;
import en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile;
import en.tresz.spaceinvaders.util.ImageLoader;
import en.tresz.spaceinvaders.util.Vector2D;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.aliens.WideAlien](#)

A wide alien that shoots three projectiles simultaneously.

Packages

- package [en.tresz.spaceinvaders.game.objects.aliens](#)

9.9 src/main/java/en/tresz/spaceinvaders/game/objects/GameObject.java File Reference

```
import en.tresz.spaceinvaders.game.GamePanel;
import en.tresz.spaceinvaders.util.Vector2D;
import java.awt.Graphics;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.GameObject](#)

The base class for all game objects.

Packages

- package [en.tresz.spaceinvaders.game.objects](#)

9.10 src/main/java/en/tresz/spaceinvaders/game/objects/Player.java File Reference

```
import en.tresz.spaceinvaders.game.GamePanel;
import en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile;
import en.tresz.spaceinvaders.util.ImageLoader;
import en.tresz.spaceinvaders.util.Vector2D;
import java.awt.Graphics;
import java.awt.image.BufferedImage;
import javax.swing.JComponent;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.Player](#)
Represents the player's spaceship controlled by keyboard input.

Packages

- package [en.tresz.spaceinvaders.game.objects](#)

9.11 src/main/java/en/tresz/spaceinvaders/game/objects/projectiles/↵ AlienProjectile.java File Reference

```
import java.awt.Graphics;
import java.awt.image.BufferedImage;
import en.tresz.spaceinvaders.game.GamePanel;
import en.tresz.spaceinvaders.game.objects.GameObject;
import en.tresz.spaceinvaders.util.ImageLoader;
import en.tresz.spaceinvaders.util.Vector2D;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile](#)
A projectile fired by aliens moving downwards.

Packages

- package [en.tresz.spaceinvaders.game.objects.projectiles](#)

9.12 src/main/java/en/tresz/spaceinvaders/game/objects/projectiles/↵ PlayerProjectile.java File Reference

```
import en.tresz.spaceinvaders.game.GamePanel;
import en.tresz.spaceinvaders.game.objects.GameObject;
import en.tresz.spaceinvaders.util.Vector2D;
import java.awt.Graphics;
import java.awt.image.BufferedImage;
```

Classes

- class [en.tresz.spaceinvaders.game.objects.projectiles.PlayerProjectile](#)
A projectile fired by the player moving upwards.

Packages

- package [en.tresz.spaceinvaders.game.objects.projectiles](#)

9.13 src/main/java/en/tresz/spaceinvaders/game/PlayerController.java File Reference

```
import en.tresz.spaceinvaders.game.objects.GameObject;  
import en.tresz.spaceinvaders.game.objects.Player;  
import en.tresz.spaceinvaders.game.objects.projectiles.AlienProjectile;
```

Classes

- class [en.tresz.spaceinvaders.game.PlayerController](#)
Handles player-related game logic including collision detection with alien projectiles.

Packages

- package [en.tresz.spaceinvaders.game](#)

9.14 src/main/java/en/tresz/spaceinvaders/GameOverWindow.java File Reference

```
import en.tresz.spaceinvaders.util.ButtonMaker.buttonSetup;  
import java.awt.*;  
import java.awt.event.*;  
import javax.swing.*;
```

Classes

- class [en.tresz.spaceinvaders.GameOverWindow](#)
Game over window for easy/medium difficulty with play again option.

Packages

- package [en.tresz.spaceinvaders](#)

9.15 src/main/java/en/tresz/spaceinvaders/HardGameOverWindow.java File Reference

```
import en.tresz.spaceinvaders.util.ButtonMaker.buttonSetup;
import java.awt.*;
import java.awt.event.*;
import javax.swing.*;
import en.tresz.spaceinvaders.util.GameTimer;
import en.tresz.spaceinvaders.util.Score;
import en.tresz.spaceinvaders.util.ScoreManager;
import en.tresz.spaceinvaders.util.Time;
```

Classes

- class [en.tresz.spaceinvaders.HardGameOverWindow](#)
Game over window for hard difficulty that allows entering player name for scoreboard.

Packages

- package [en.tresz.spaceinvaders](#)

9.16 src/main/java/en/tresz/spaceinvaders/Main.java File Reference

Classes

- class [en.tresz.spaceinvaders.Main](#)
The entry point of the Space Invaders application.

Packages

- package [en.tresz.spaceinvaders](#)

9.17 src/main/java/en/tresz/spaceinvaders/MainWindow.java File Reference

```
import en.tresz.spaceinvaders.game.GamePanel;
import en.tresz.spaceinvaders.menu.DifficultyMenuPanel;
import en.tresz.spaceinvaders.menu.MainMenuPanel;
import en.tresz.spaceinvaders.menu.ScoreboardPanel;
import javax.swing.JFrame;
import javax.swing.JPanel;
import javax.swing.WindowConstants;
import java.awt.CardLayout;
```

Classes

- class [en.tresz.spaceinvaders.MainWindow](#)
The main frame.
- enum [en.tresz.spaceinvaders.MainWindow.PanelType](#)

Packages

- package [en.tresz.spaceinvaders](#)

9.18 src/main/java/en/tresz/spaceinvaders/menu/DifficultyMenuPanel.java File Reference

```
import en.tresz.spaceinvaders.MainWindow;
import en.tresz.spaceinvaders.util.ButtonMaker;
import en.tresz.spaceinvaders.util.ImageLoader;
import javax.swing.JButton;
import javax.swing.JPanel;
import java.awt.GridBagLayout;
import java.awt.GridBagConstraints;
import java.awt.Insets;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.awt.image.BufferedImage;
import java.awt.Graphics;
```

Classes

- class [en.tresz.spaceinvaders.menu.DifficultyMenuPanel](#)
The difficulty selection screen.

Packages

- package [en.tresz.spaceinvaders.menu](#)

9.19 src/main/java/en/tresz/spaceinvaders/menu/MainMenuPanel.java File Reference

```
import en.tresz.spaceinvaders.MainWindow;
import en.tresz.spaceinvaders.util.ButtonMaker;
import en.tresz.spaceinvaders.util.ImageLoader;
import javax.swing.Box;
import javax.swing.JButton;
import javax.swing.JLabel;
import javax.swing.JOptionPane;
import javax.swing.JPanel;
import javax.swing.SwingUtilities;
```

```
import java.awt.GridBagLayout;
import java.awt.GridBagConstraints;
import java.awt.Insets;
import java.awt.event.ActionListener;
import java.awt.event.ActionEvent;
import java.awt.Graphics;
import java.awt.BorderLayout;
import java.awt.Window;
import java.awt.image.BufferedImage;
```

Classes

- class [en.tresz.spaceinvaders.menu.MainMenuPanel](#)
The main menu panel with options to play, view scoreboard, or exit.

Packages

- package [en.tresz.spaceinvaders.menu](#)

9.20 src/main/java/en/tresz/spaceinvaders/menu/ScoreboardPanel.java File Reference

```
import en.tresz.spaceinvaders.MainWindow;
import en.tresz.spaceinvaders.util.ButtonMaker;
import en.tresz.spaceinvaders.util.Score;
import en.tresz.spaceinvaders.util.ScoreManager;
import javax.swing.JButton;
import javax.swing.JPanel;
import javax.swing.JScrollPane;
import javax.swing.JLabel;
import javax.swing.BorderFactory;
import javax.swing.text.SimpleAttributeSet;
import javax.swing.text.StyleConstants;
import javax.swing.text.StyledDocument;
import javax.swing.JTextPane;
import java.awt.BorderLayout;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
import java.util.List;
```

Classes

- class [en.tresz.spaceinvaders.menu.ScoreboardPanel](#)
The scoreboard panel displays the top 5 high scores.

Packages

- package [en.tresz.spaceinvaders.menu](#)

9.21 src/main/java/en/tresz/spaceinvaders/util/ButtonMaker.java File Reference

```
import javax.swing.ImageIcon;
import javax.swing.JButton;
```

Classes

- class [en.tresz.spaceinvaders.util.ButtonMaker](#)
A utility class to create customized buttons.

Packages

- package [en.tresz.spaceinvaders.util](#)

9.22 src/main/java/en/tresz/spaceinvaders/util/GameTimer.java File Reference

```
import javax.swing.*;
import java.awt.*;
```

Classes

- class [en.tresz.spaceinvaders.util.GameTimer](#)
A label that displays and tracks game time in MM:SS format.

Packages

- package [en.tresz.spaceinvaders.util](#)

9.23 src/main/java/en/tresz/spaceinvaders/util/HealthBar.java File Reference

```
import javax.swing.*;
import java.awt.*;
import java.awt.image.BufferedImage;
```

Classes

- class [en.tresz.spaceinvaders.util.HealthBar](#)
A visual health bar component that displays health as heart images.

Packages

- package [en.tresz.spaceinvaders.util](#)

9.24 src/main/java/en/tresz/spaceinvaders/util/ImageLoader.java File Reference

```
import java.net.URL;
import javax.imageio.ImageIO;
import javax.swing.ImageIcon;
import java.awt.image.BufferedImage;
import java.io.IOException;
```

Classes

- class [en.tresz.spaceinvaders.util.ImageLoader](#)
A utility class to safely load images from the resources folder.

Packages

- package [en.tresz.spaceinvaders.util](#)

9.25 src/main/java/en/tresz/spaceinvaders/util/Score.java File Reference

Classes

- class [en.tresz.spaceinvaders.util.Score](#)
Represents a player's score with their name and completion time.

Packages

- package [en.tresz.spaceinvaders.util](#)

9.26 src/main/java/en/tresz/spaceinvaders/util/ScoreManager.java File Reference

```
import java.io.File;
import java.io.FileReader;
import java.io.FileWriter;
import java.io.IOException;
import java.io.Reader;
import java.io.Writer;
import java.lang.reflect.Type;
import java.util.ArrayList;
import java.util.List;
import com.google.gson.Gson;
import com.google.gson.GsonBuilder;
import com.google.gson.reflect.TypeToken;
```

Classes

- class [en.tresz.spaceinvaders.util.ScoreManager](#)
Manages saving and loading of high scores using JSON serialization.

Packages

- package [en.tresz.spaceinvaders.util](#)

9.27 src/main/java/en/tresz/spaceinvaders/util/Time.java File Reference

Classes

- class [en.tresz.spaceinvaders.util.Time](#)
Represents a time duration in minutes and seconds.

Packages

- package [en.tresz.spaceinvaders.util](#)

9.28 src/main/java/en/tresz/spaceinvaders/util/Vector2D.java File Reference

Classes

- class [en.tresz.spaceinvaders.util.Vector2D](#)
2D vector class for positioning.

Packages

- package [en.tresz.spaceinvaders.util](#)

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